arXiv:2603.07980v2 [cs.LG] 8 Oct 2026

$OneMillion-Bench: How Far are Language Agents from Human Experts?

Yang Liu[2⋆], **Jiaqi Li**[2⋆], **Jun Bai**[2⋆], **Qianyu Yang**[1⋆], **Xiaobo Hu**[3], **Tao Peng**[1], **Zaiyuan Wang**[1], **Ran Tian**[1], **Jiayun Dong**[1], **Chun Zhang**[1], **Zixia Jia**[2], **Kaiyuan Chen**[3], **Yixin Ren**[3], **Yang Liu**[3], **Yanglihong Xiao**[1], **Lingyue Yin**[1], **Tiliang Duan**[1], **Ge Zhang**[4], **Gang Yao**[2], **Hao Chen**[2], **Yuan Gong**[3†], **Jianpeng Jiao**[1†], **Zilong Zheng**[2†]

[1]**Humanlaya**, [2]**BIGAI**, [3]**xbench**, [4]**M-A-P**

Abstract

As language models (LMs) evolve from chat assistants to long-horizon agents capable of multi-step reasoning and tool use, existing benchmarks remain largely confined to structured or exam-style tasks that fall short of real-world professional demands. To this end, we introduce **$OneMillion-Bench** ($OMB), a benchmark of 400 expert-curated tasks spanning Law, Finance, Industry, Healthcare, and Natural Science, built to evaluate agents across economically consequential scenarios. Unlike prior work, the benchmark requires retrieving authoritative sources, resolving conflicting evidence, applying domain-specific rules, and making constraint decisions, where correctness depends as much on the reasoning process as the final answer. We adopt a rubric-based evaluation protocol scoring factual accuracy, logical coherence, practical feasibility, and professional compliance, focusing on expert-level problems to ensure meaningful differentiation across agents. Together, $OMB provides a unified testbed for assessing agentic reliability, professional depth, and an indicator of practical readiness in domain-intensive scenarios.

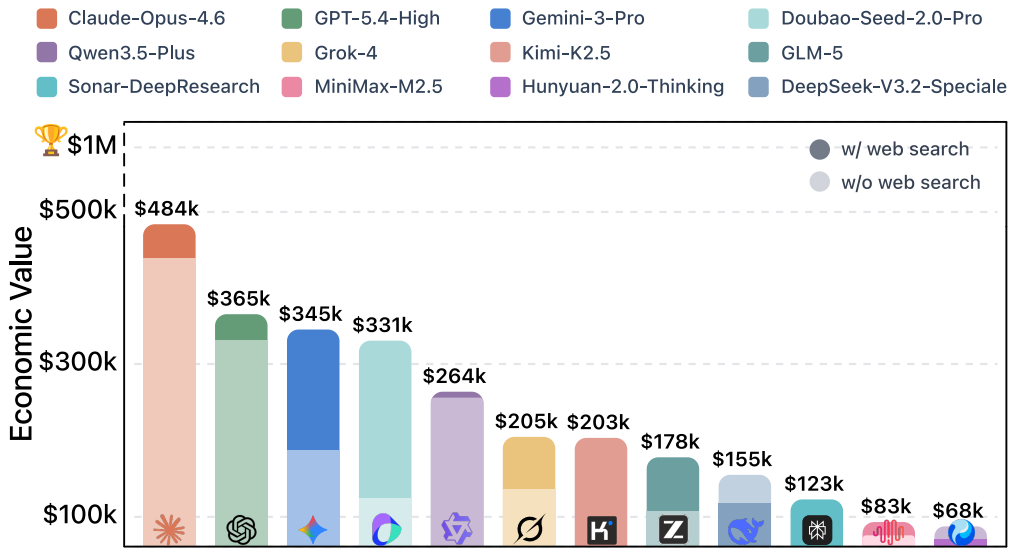


Figure 1: Leaderboard performance on $OneMillion-Bench.

⋆Co-first Authors. † Corresponding Authors. Complete list of dataset contributors is in Appendix A. Correspondence to jasper@humanlaya.com and zlzheng@bigai.ai.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Track on Evaluations and Datasets.

1 Introduction

Consider work performed daily in professional industries: a senior actuary auditing a life-insurance reserve valuation under IFRS 17; an M&A lawyer verifying cross-border compliance clauses; or an investment analyst constructing a high-pressure valuation model. These workloads are not merely prompts to be answered, but context-heavy professional deliverables that require specialized knowledge, long-horizon execution, and strict adherence to domain constraints. While language agents are capable of supporting complex reasoning [11, 17, 30, 18, 6], multi-turn interactions [42, 3, 41], and realistic decision making [2, 45], a critical gap remains between exam-style evaluations and the rigorous demands of professional labor. Existing benchmarks evaluate agents on solving isolated questions but lack clarity and provide limited insight into whether they can perform professional work and reliably create value in economically consequential professional scenarios [33, 10, 16].

To effectively quantify the economic value that language agents can deliver in realistic tasks, we create **$OneMillion-Bench**, a comprehensive benchmark built around expert tasks designed to measure not only precision, but value. It spans high-impact professional domains, requires intensive retrieval, evidence-based reasoning, and tests reliability under explicit constraints. The benchmark comprises 400 challenging open-ended tasks in five domains: Finance, Law, Healthcare, Natural Science, and Industry, curated through a collaborative effort spanning over 2,000 expert hours. Each task is assigned a realistic monetary value, calculated by the estimated time required for the task completion by a senior professional and its prevailing market hourly wage[1][2]. The estimated value of all tasks exceeds $1 million, giving the benchmark its name. Performance on $OneMillion-Bench is thus expressed by translating agent capability directly into a value measure of professional utility.

$OneMillion-Bench offers several key features: (1) **Economic evaluation** that quantifies agentic capabilities through the lens of labour costs. It distinguishes the capabilities of language agents through how much they earn in million-value professional tasks. (2) **Comprehensive domain coverage** spans five professional areas, with balanced sub-domain partitioning. (3) **Skill-oriented taxonomy** including retrieval, reasoning, verbalization, and instruction following that enables fine-grained analysis, instead of overfitting evaluation to any domain idiosyncrasy. (4) **Rubric-based evaluation** mechanism avoids evaluation metric gaming [38] and well aligns with domain-specific policies and expectations designed by domain experts to provide fine-grained automated scoring.

Ultimately, $OneMillion-Bench is a step towards value quantification of language agents in real-world professional scenarios which turns capability into an interpretable quantity: how much reliable work an agent can actually deliver, and how much that work is worth, accelerating progress towards AI systems that are *not only powerful, but trustworthy and economically meaningful*. **We release all the artifacts of $OneMillion-Bench in** https://github.com/humanlaya/OneMillion-Bench**.**

2 Constructing $OneMillion-Bench

2.1 Data Curation Pipeline

Following the principle: (1) realistic workflow problems with high economic value; (2) tasks with objective and quantifiable rubrics; (3) diverse rubric structures without metric gaming, the data curation follows a rigorous three-stage expert annotation process to ensure objectivity and reliability in Figure 2.

Stage 1: Task Creation. An expert with professional domain knowledge begins by designing a specialized task, which includes: (1) formulating a specific and valuable task; (2) drafting a comprehensive reference answer that illustrates the underlying rationale; and (3) defining detailed scoring rubrics. These tasks are semi-open-ended, allowing different valid approaches, and are time-agnostic, yielding stable benchmarks over time. A critical subsequent step is concurrent adversarial validation. The task is tested against state-of-the-art agents from different model families to minimize bias, and retained only if several agents fail to meet a predefined passing threshold based on rubrics.

Stage 2: Peer Review. After the initial draft is completed, the task undergoes an independent peer review by a second expert in the same sub-domain. The reviewer conducts a holistic evaluation of the

[1] https://data.bls.gov/oes/#/industry/000000

[2] https://rsj.gz.gov.cn/zwdt/tzgg/content/post_10619688.html

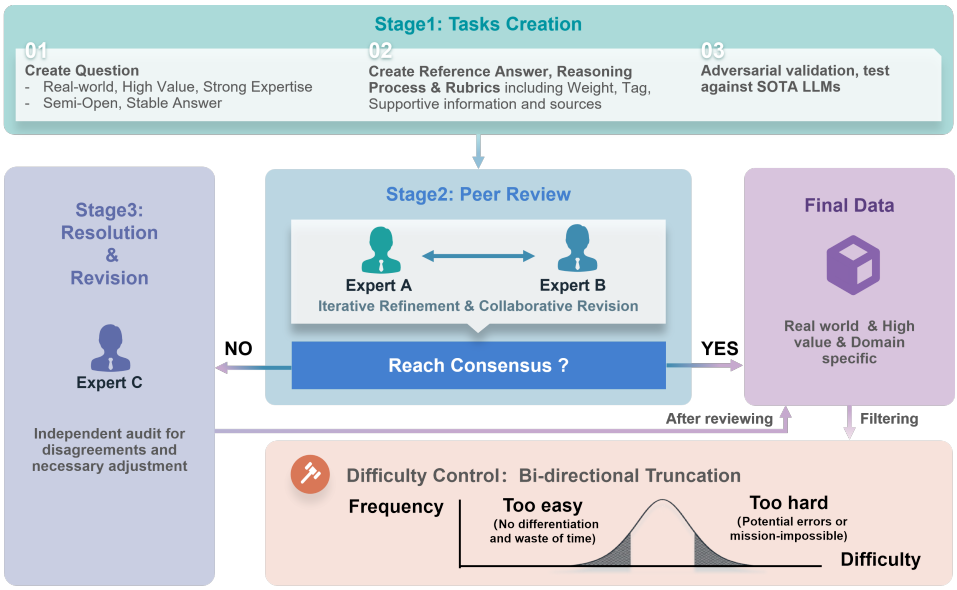


Figure 2: Data curation pipeline of $OneMillion-Bench.

task's clarity, domain specialization, adherence to design principles, and objectivity of the rubrics. Structured feedback leads to iterative discussions and collaborative revisions between the two experts, continuing until they reach a final agreement on the task, the reference response, and the evaluation criteria.

Stage 3: Resolution and Revision. For cases exhibiting potential risk or instances where the first two experts can not achieve consensus, a third expert performs an independent audit. This final stage involves necessary adjustments to guarantee the high quality and reliability of the curated data.

To enhance the overall discriminability, we further filtered the candidate dataset with bidirectional truncation by removing or revising tasks at both extremes:

- **Lower Bound Elimination:** Tasks with excessively low difficulty are removed. They are solved by all the agents tested that are deemed a waste of evaluation resources without differentiation.

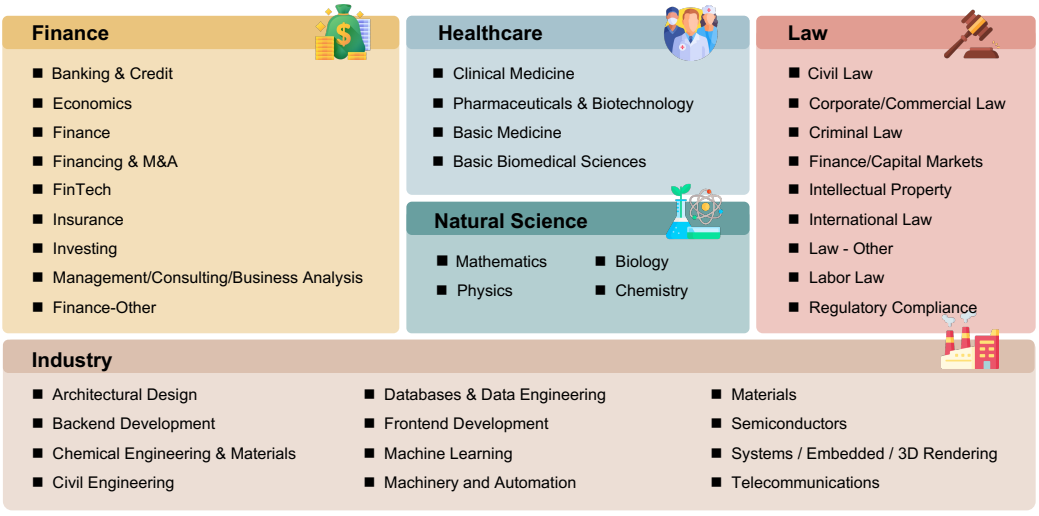


Figure 3: $OneMillion-Bench consists of 5 macro domains, 37 sub-domains and 92 third-level categories covering a wide variety of real applications and professional scenarios (*cf.* Appendix C).

- **Upper Bound Review:** Tasks with extreme difficulty, where agents consistently yield universally low scores, undergo a secondary review to avoid unanswerable or infeasible tasks.

Table 1: Data Statistics of $OneMillion-Bench

Subset	Domain	# Questions	Question Len.	# Rubrics	# Negative Rubrics
CN	Finance	40	554.8	18.8	4.2
	Law	40	709.1	16.3	3.9
	Healthcare	40	584.2	16.9	4.3
	Natural Science	40	555.8	16.8	4.3
	Industry	40	965.2	15.9	4.0
Global	Finance	40	1810.9	18.5	4.1
	Law	40	3277.8	16.0	4.2
	Healthcare	40	2029.9	16.9	4.2
	Natural Science	40	1634.6	16.8	4.3
	Industry	40	2581.1	15.9	4.0

2.2 Data Overview

Comprehensive Domain Coverage. $OneMillion-Bench comprises a meticulously curated collection of 400 questions categorized into 5 domains, including law, finance, healthcare, natural science and industry (Table 1 and Figure 3). These questions are tailored to the actual needs of different positions, workflows and constraints, which is critical for testing agents in a wide array of real-world scenarios, thereby facilitating a comprehensive and robust evaluation. Examples are depicted in Figure 7 for reference.

Workflow-Realistic Difficulty and High Economic Value Problems. The benchmark focuses on developing real-world agentic tasks across various vertical domains, ensuring that the task design closely matches the actual needs and workflows of the industry, thereby effectively evaluating the decision-making and task execution capabilities of agents in complex professional environments.

$OneMillion-Bench tasks are constructed to simulate professional workflows which not only require the agent to perform deep reasoning and real-time information integration, but also to clearly distinguish the practical capabilities and economic value of different agents through their judgments and insights. Unlike the exam-style questions of existing benchmarks [32, 30], many tasks in $OneMillion-Bench are intentionally structured so that a correct final answer is inseparable from a correct process: precise retrieval, traceable justification, and careful constraint satisfaction.

Rubrics-based Grading with Negative Penalty. Rubrics in $OneMillion-Bench are primarily determined case-by-case across domains and hand-crafted by domain experts. It ensures the rubrics are tailored and precise, moving beyond generic scoring to incorporate domain-specific logic, factual accuracy, and analytical depth. Each effective rubric evaluation consists of four fundamental elements:

- **Rubrics:** Evaluation criteria, traits, and dimensions that determine response correctness.
- **Rubric Weights:** Gains or penalties reflecting the relative importance of each rubric.
- **Rubric Tags:** Categories classifying criteria by required capability.
- **Sources & Citations:** References to verified external sources for factual accuracy.

It is worth highlighting that we propose a **negative rubric scoring** mechanism with asymmetric rubric weights ranging from -20 to 10 to better align with realistic domain requirements. The negative rubrics are mostly around behavioral violations of industrial norms or professional conduct, harmful generation, hallucination, and lapses in expected competencies such as instruction following.

Diverse Agentic Capabilities Assessment. For each rubric, we associate a rubric tag indicating the agentic capability required to finish the task. **(1) Web Search:** This evaluates the agentic ability to locate and extract factual information, entity facts, and authoritative definitions from retrieved sources. It primarily assesses the information retrieval capability, focusing on questions of "what". **(2) Reasoning:** This aspect examines how agents make causal attributions, coherent trend judgments,

Table 2: Comparison of Economic Value ($USD), Expert Score (%), and Pass Rate(%) on **Global set**. The sign of 🌐 denotes enabling the usage of web search tool, and the same holds for Table 3 and 4.

Model / Agentic System	Economic Value		Expert Score		Pass Rate	
	Vanilla	w/ 🌐	Vanilla	w/ 🌐	Vanilla	w/ 🌐
CLAUDE-OPUS-4.6	439.2k	483.8k ↑44.6	55.0	63.0 ↑8.1	36.5	43.5 ↑7.0
GPT-5.4-HIGH	330.9k	365.5k ↑34.6	55.0	59.2 ↑4.2	31.5	38.0 ↑6.5
QWEN3.5-PLUS	256.0k	263.7k ↑7.8	47.6	49.0 ↑1.4	22.5	23.5 ↑1.0
GPT-5.2-HIGH	236.6k	335.9k ↑99.3	47.6	56.9 ↑9.3	23.0	32.0 ↑9.0
KIMI-K2.5	203.3k	162.7k ↓40.6	45.1	41.2 ↓3.9	17.5	17.5 ↑0.0
GPT-5.3-CODEX	201.1k	317.8k ↑116.7	48.1	50.6 ↑2.6	18.5	29.0 ↑10.5
GEMINI-3-PRO-PREVIEW	187.3k	345.3k ↑158.0	43.7	52.8 ↑9.1	15.5	28.5 ↑13.0
DEEPSEEK-V3.2-SPECIALE	154.9k	118.0k ↓37.0	40.5	38.6 ↓1.8	16.0	13.0 ↓3.0
GROK-4	136.3k	204.6k ↑68.3	38.6	45.9 ↑7.2	12.0	18.5 ↑6.5
DOUBAO-SEED-2.0-PRO	123.9k	330.7k ↑206.9	39.4	51.8 ↑12.4	13.5	28.5 ↑15.0
GLM-5	107.8k	177.9k ↑70.1	38.5	41.1 ↑2.6	9.5	16.5 ↑7.0
MINIMAX-M2.1	85.6k	46.6k ↓39.0	34.9	33.9 ↓1.0	8.0	5.5 ↓2.5
LING-2.5-1T	79.5k	50.0k ↓29.5	33.1	31.2 ↓1.9	7.5	6.5 ↓1.0
STEP-3.5-FLASH	68.6k	46.9k ↓21.7	34.4	33.0 ↓1.4	7.0	5.0 ↓2.0
HUNYUAN-2.0-THINKING	68.0k	25.6k ↓42.3	34.7	30.2 ↓4.5	8.5	3.0 ↓5.5
MINIMAX-M2.5	37.9k	82.6k ↑44.8	29.5	31.5 ↑2.0	4.0	8.0 ↑4.0
Deep Research Agents						
O3-DEEPRESEARCH	181.7k		46.3		17.0	
SONAR-DEEPRESEARCH	122.8k		42.3		13.5	
O4-MINI-DEEPRESEARCH	112.6k		37.7		10.5	

and logical deductions, addressing questions of "why" and "how". **(3) Verbalization:** This evaluates the organization, language style, and readability of the generation, ensuring clear, professional, and appropriate communication. **(4) Instruction Following:** Compliance with instructions is critical for reliability in professional tasks. This assesses the adherence of agents to explicit constraints, rules, and guidance, ensuring controllable, predictable behaviors that meet user requirements.

Bilinguality Integrated with Local Culture. Although there are prior general-purpose agentic benchmarks (*e.g.*, GDPVal [27]), they operate under a default English or western-centric context, failing to capture significant variations in tasks, regulations, and practical scenarios encountered in different linguistic and cultural environments. Therefore, $OneMillion-Bench consists of 200 English and 200 Chinese instances. The Chinese (CN) set, however, is not a direct translation in parallelism to the Global set, but a purpose-built collection that incorporates tasks specific to the context of the China Mainland. This includes local regulations and industrial standards (*e.g.*, Chinese Cybersecurity Law, healthcare insurance policies, and accounting standards) and cultural nuances in problems.

3 Benchmarking Frontier Agents on $OneMillion-Bench

3.1 Evaluated Systems

In this section, we present the evaluation results for $OneMillion-Bench, encompassing a total of 37 models and agents, which are grouped into three categories according to their characteristics:

- **Vanilla Models (17 models):** The vanilla models represent open-source and proprietary models evaluated without tool usage, including Claude-Opus-4.6 [1], DeepSeek-V3.2-Speciale [13], Doubao-Seed-2.0-Pro [5], Gemini-3-Pro-Preview [8], Gemini-3.1-Pro-Preview[37], GLM-5 [44], GPT-5.2-High [24], GPT-5.3-Codex [25], GPT-5.4-High [26], Grok-4 [40], Hunyuan-2.0-Thinking [36], Kimi-K2.5 [21], Ling-2.5-1T [9], MiniMax-M2.1 [20], MiniMax-M2.5 [19], Qwen3.5-Plus [31], and Step-3.5-Flash [35].
- **Web Search Agents (vanilla models equipped with web search tools, 17 agents):** This category evaluates vanilla models listed above equipped with web search tool calling for external information retrieval, *e.g.*, Claude-Opus-4.6 (Web Search) [1].

Table 3: Comparison of Economic Value (¥CNY), Expert Score (%), and Pass Rate (%) on **CN set**.

Model / Agentic System	Economic Value		Expert Score		Pass Rate	
	Vanilla	w/ 🌐	Vanilla	w/ 🌐	Vanilla	w/ 🌐
CLAUDE-OPUS-4.6	350.3k	470.2k ↑119.9	55.8	64.5 ↑8.7	35.0	48.5 ↑13.5
GPT-5.4-HIGH	339.5k	305.4k ↓34.1	56.2	58.7 ↑2.5	35.5	35.5 ↑0.0
QWEN3.5-PLUS	304.9k	308.1k ↑3.2	51.4	52.1 ↑0.6	28.0	28.0 ↑0.0
GPT-5.2-HIGH	226.9k	294.0k ↑67.1	47.1	52.6 ↑5.5	22.0	28.0 ↑6.0
GEMINI-3-PRO-PREVIEW	202.9k	332.5k ↑129.7	46.0	54.0 ↑8.0	20.0	30.0 ↑10.0
KIMI-K2.5	196.5k	126.0k ↓70.4	45.4	43.7 ↓1.7	19.5	15.0 ↓4.5
GPT-5.3-CODEX	178.4k	201.1k ↑22.7	46.9	50.9 ↑4.1	19.0	23.0 ↑4.0
DEEPSEEK-V3.2-SPECIALE	139.4k	155.8k ↑16.4	41.1	44.4 ↑3.3	14.0	16.5 ↑2.5
GLM-5	138.9k	136.3k ↓2.6	40.7	41.1 ↑0.4	13.0	11.0 ↓2.0
LING-2.5-1T	119.8k	101.8k ↓17.9	36.7	35.6 ↓1.1	11.0	11.5 ↑0.5
DOUBAO-SEED-2.0-PRO	110.8k	255.9k ↑145.1	40.6	50.5 ↑9.8	11.5	25.0 ↑13.5
MINIMAX-M2.1	106.6k	67.2k ↓39.4	35.8	36.7 ↑1.0	10.0	6.0 ↓4.0
GROK-4	104.2k	99.4k ↓4.8	39.8	39.6 ↓0.2	11.0	11.5 ↑0.5
HUNYUAN-2.0-THINKING	83.0k	53.8k ↓29.2	36.3	32.8 ↓3.4	9.5	6.5 ↓3.0
MINIMAX-M2.5	70.3k	51.7k ↓18.7	33.4	32.1 ↓1.3	7.5	7.5 ↑0.0
STEP-3.5-FLASH	52.6k	68.7k ↑16.2	36.2	35.6 ↓0.6	6.5	6.5 ↑0.0
Deep Research Agents						
O3-DEEPRESEARCH	149.2k		41.7		16.0	
SONAR-DEEPRESEARCH	129.5k		40.2		13.5	
O4-MINI-DEEPRESEARCH	84.7k		36.0		9.0	

- **Deep Research Agents (3 agents):** These are specialized systems for long-horizon research, including o3-DeepResearch [23], o4-Mini-DeepResearch [23], and Sonar-DeepResearch [29].

3.2 Evaluation Metrics

Expert Score. It is introduced to assess the degree to which model generations fulfill professional criteria across all questions. For a question set Q, the normalized Expert Score is defined as

$$\text{Expert Score}(Q) = \frac{1}{|Q|} \sum_{q \in Q} \max\left(0, \frac{\sum_{r \in R_q} s_r}{\sum_{r \in R_q^+} w_r}\right), \tag{1}$$

where R_q denotes the rubric set for question q, R_q^+ the subset with positive weights, s_r the score of rubric r, and w_r its predefined weight. Question scores are clipped to $[0, 1]$ before aggregation.

Pass Rate. We also define Pass Rate, a binary success metric indicating whether a generation meets a minimum professional standard via calculating the average Expert Score across all questions Q:

$$\text{Pass Rate}(Q) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}\left[\text{Expert Score}(q) \geq 0.7\right], \tag{2}$$

where $\mathbb{1}[\cdot]$ denotes the indicator function based on the given binary scoring criterion.

Economic Value. The economic value of an agent is the cumulative value of all passed questions:

$$\text{Economic Value}(Q) = \sum_{q \in Q} V_q \cdot \mathbb{1}\left[\text{Expert Score}(q) \geq 0.7\right], \tag{3}$$

where V_q denotes the estimated economic value of question q. **Specifically, we provide a detailed approach and guideline of reproducibility for economic value calculation in Appendix B.** We follow Dou et al. [7] to use 0.7 as the cliff threshold in our default setting.

Table 4: Expert Score aggregated by rubric type, averaged over Global and CN subsets. FI: Factual Information. AR: Analytical Reasoning. IF: Instructions Following. SF: Structure and Formatting.

Model / Agentic System	FI		AR		IF		SF	
	Vanilla	w/ 🌐	Vanilla	w/ 🌐	Vanilla	w/ 🌐	Vanilla	w/ 🌐
GPT-5.2-HIGH	47.4	59.4 ↑12.0	52.3	56.4 ↑4.1	66.0	72.1 ↑6.1	82.7	85.6 ↑2.9
QWEN3.5-PLUS	50.3	50.0 ↓0.3	58.9	59.9 ↑1.0	73.2	74.6 ↑1.4	86.4	88.8 ↑2.4
CLAUDE-OPUS-4.6	54.0	66.5 ↑12.5	61.3	73.0 ↑11.7	74.6	82.1 ↑7.5	88.9	90.0 ↑1.1
GPT-5.4-HIGH	53.5	63.5 ↑10.0	60.8	61.6 ↑0.8	71.5	73.9 ↑2.4	86.4	84.5 ↓1.9
GEMINI-3.1-PRO-PREVIEW	49.8	46.7 ↓3.1	51.9	50.8 ↓1.1	70.6	67.9 ↓2.7	82.7	82.7 ↑0.0
GPT-5.3-CODEX	48.5	58.4 ↑9.9	51.1	52.0 ↑0.9	67.5	65.9 ↓1.6	85.0	83.6 ↓1.4
KIMI-K2.5	46.5	44.1 ↓2.4	51.1	48.1 ↓3.0	70.1	63.4 ↓6.7	85.0	80.6 ↓4.4
GEMINI-3-PRO-PREVIEW	45.8	56.7 ↑10.9	49.2	60.7 ↑11.5	66.9	73.1 ↑6.2	82.8	86.8 ↑4.0
GROK-4	41.6	48.5 ↑6.9	44.7	46.7 ↑2.0	65.2	62.7 ↓2.5	83.0	86.5 ↑3.5
DOUBAO-SEED-2.0-PRO	42.3	53.6 ↑11.3	45.3	58.8 ↑13.5	62.3	72.7 ↑10.4	83.6	86.8 ↑3.2
DEEPSEEK-V3.2-SPECIALE	43.7	43.9 ↑0.2	45.4	46.8 ↑1.4	62.8	60.2 ↓2.6	79.4	78.8 ↓0.6
GLM-5	41.8	40.8 ↓1.0	45.1	47.7 ↑2.6	65.3	65.7 ↑0.4	79.4	82.7 ↑3.3
HUNYUAN-2.0-THINKING	39.1	36.3 ↓2.8	40.2	36.5 ↓3.7	63.5	60.9 ↓2.6	85.0	79.8 ↓5.2
MINIMAX-M2.1	39.3	38.3 ↓1.0	41.6	42.4 ↑0.8	60.4	57.4 ↓3.0	82.6	81.8 ↓0.8
STEP-3.5-FLASH	36.2	38.4 ↑2.2	42.5	41.6 ↓0.9	62.3	56.1 ↓6.2	82.6	79.2 ↓3.4
LING-2.5-1T	37.6	36.5 ↓1.1	41.6	38.7 ↓2.9	59.3	55.9 ↓3.4	79.3	74.4 ↓4.9
MINIMAX-M2.5	34.9	35.3 ↑0.4	39.1	35.8 ↓3.3	58.2	55.4 ↓2.8	83.0	71.5 ↓11.5
Deep Research Agents								
SONAR-DEEPRESEARCH	47.2		49.9		60.7		80.7	
O3-DEEPRESEARCH	47.9		50.2		59.2		76.5	
O4-MINI-DEEPRESEARCH	42.9		42.3		58.1		71.7	

3.3 Main Results

Tables 2, 3, 8, and 9 present the comparison for agents evaluated in the Global and CN subsets, respectively. From these empirical results, several key observations emerge and are listed below.

The Strong Get Stronger: Web Search Rewards Capable Models. In both Global and CN subsets, CLAUDE-OPUS-4.6 leads among vanilla models and further extends its advantage when web search is enabled, suggesting that it mainly benefits models capable of converting retrieved evidence into reliable decisions. Stronger models excel at forming precise queries, identifying authoritative sources, resolving conflicts, and integrating the retrieved facts with task constraints. Weaker models tend to treat search results as raw context and are more easily misled by noisy evidence. For example, HUNYUAN-2.0 and STEP-3.5-FLASH shows similar declines after enabling web search, indicating that retrieval can hurt performance when models lack the ability to select and reconcile evidence.

Partial Credit, Low Delivery: The Gap Between Scoring and Solving. Although many agents reach moderate Expert Scores (*e.g.*, 45%–50%), their Pass Rates often remain low (below 25%), indicating partial rubric satisfaction rather than full task competency. For instance, in Finance, an agent may extract relevant figures and give a plausible judgment, yet fail by missing a required calculation, using an outdated threshold, or overlooking a key exception. This also explains why Finance remains harder than others for top models: it requires exact numerical reasoning and strict constraint checking, where incomplete solutions receive limited credit, suggesting that difficulty reflects both rubric fullfillness demanding strong domain knowledge.

Global *vs*. CN Differences Reveal Region-Specific Generalization Gaps. Global and CN results expose distinct capability bottlenecks across agents. Top-tier agents remain relatively stable: CLAUDE-OPUS-4.6 obtains similar vanilla Expert Scores on Global and CN (55.0% *vs*. 55.8%), as does GPT-5.4-HIGH (55.0% *vs*. 56.2%). This suggests that their performance relies more on general evidence understanding and rubric-aligned execution than on a specific regional distribution. By contrast, QWEN3.5-PLUS performs better on CN than Global (51.4% *vs*. 47.6%), likely due to stronger coverage of Chinese knowledge and evidence. Overall, cross-region robustness depends not only on agent strength, but also on adapting reasoning to different evidence contexts.

4 Analyses

4.1 Performance Analyses across Rubric Type

To evaluate task performance in detail, we report the per-type scores in Table 4 and observe that:

A pronounced gap exists between superficial and evidence-centric capabilities: agents consistently outperform in *Structure and Formatting* (SF) and *Instruction Following* (IF) compared to *Factual Information* (FI) and *Analytical Reasoning* (AR), indicating agents can produce well-organized, instruction-compliant solutions while failing to deliver accurate evidence or complete analysis. Web search primarily enhances evidence-centric dimensions for frontier models, notably CLAUDE-OPUS-4.6 (+12.5% FI, +11.7% AR) and GPT-5.2-HIGH (+12.0% FI, +4.1% AR), while potentially misleading weaker models through noisy retrieval and hallucination amplification.

Web search also tests whether agents stay anchored to the original task, making IF highly dependent on specific models ranging from +10.4% for DOUBAO-SEED-2.0-PRO to -6.7% for KIMI-K2.5, as some agents shift toward summarizing retrieved content. SF shows little room for improvement and may even weaken as search encourages longer and evidence-heavy responses.

4.2 Web Search Scaffold Does Matter

To mitigate non-transferability issues in the evaluation and provide insights into the impact of agent scaffold, we investigate three setups: (1) scaffold of official provider (*e.g.*, OpenAI and Anthropic), (2) OpenRouter[3], and (3) no scaffold (w/o Web Search). In Figure 4, official scaffold outperforms OpenRouter in almost all evaluated models. The performance gap between official and OpenRouter scaffold varies considerably across models. In some cases (*e.g.*, Claude-Opus-4.6), the "w/o Web Search" baseline surprisingly surpasses the OpenRouter setup, suggesting that scaffold quality matters more than access to web search alone.

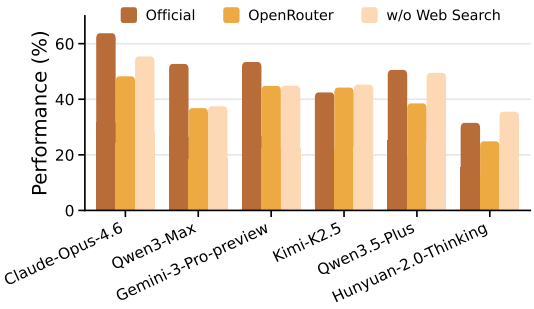


Figure 4: Comparison of web search scaffolds.

4.3 Agents Struggle for Consistent Delivery

We report pass@k and pass^k scores with performing parallel test-time scaling experiments on the Finance domain of the Global subset [34]. Overall, Figure 5 illustrates the test-time scalability of various models on $OneMillion-Bench. As k increases, the two metrics diverge sharply: pass@k (dotted lines) shows consistent logarithmic gains across all models, with Claude-Opus-4.6 leading and plateauing near 30%, while pass^k (solid lines) decays towards zero. While larger sample scales increase the probability of capturing a correct solution, the accompanying rise in output variance degrades reliability, causing consistency to collapse in large-k regimes.

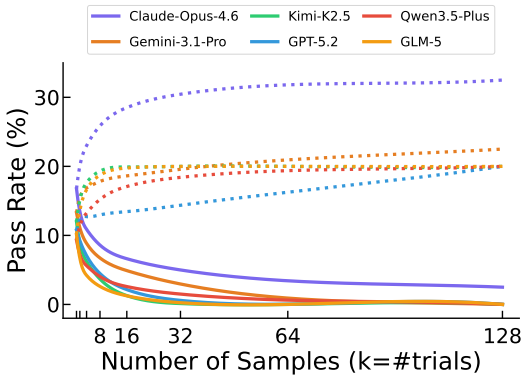


Figure 5: Test-time scalibility on Finance of the Global subset for consistency validation.

4.4 Pareto Optimality of Return on Cost

As depicted in Figure 6, agents or models achieve a pareto-optimal trade-off tendency between the outcome economic value versus the inference cost of agents or models. Specialized web search agents deliver drastically higher economic value than that of the same base model. We also notice

[3] https://openrouter.ai

that smaller models deliver near-frontier deep research performance at a fraction of the cost of the largest top-tier models, making advanced research tools accessible to price-sensitive users in the real scenarios. Based on the further case studies in Section 4.5, we find that higher token cost does not necessarily contribute to more consistent delivery in those domain specific tasks.

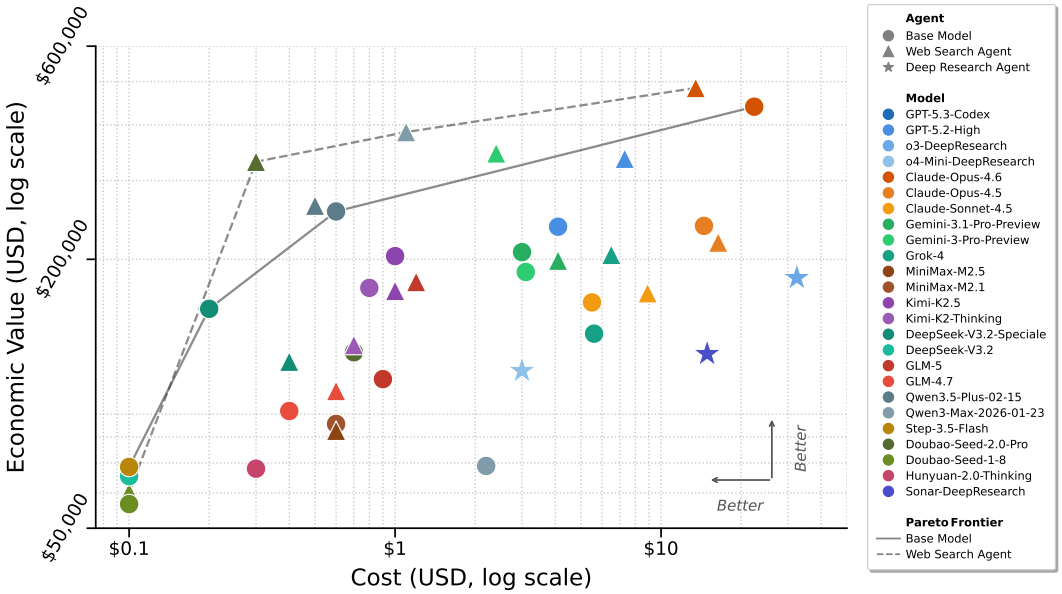


Figure 6: Pareto frontiers of various agents and models on $OneMillion-Bench.

4.5 Deficiency Analysis

We randomly choose 100 trajectories of solutions from different domains for three representative models (GPT-5.2-High, Gemini-3-Pro-preview, and Claude-Opus-4.5). Our qualitative analysis reveals several recurring failure modes in $OneMillion-Bench as follows:

Flawed Reasoning with Weak Verification (34% of Errors). The model may arrive at a plausible conclusion while failing to validate it against task-specific constraints or edge cases. A common manifestation is the omission of fine-grained rubric requirements that are essential for correctness, such as downstream demand structure, non-applicability of special categories, second-order behavioral changes, quantitative comparison against core business volatility, or explicit compliance preconditions. In this way, the final answer often does not satisfy the decisive conditions of the task. We also find insufficient boundary checking: the model may infer a threshold or cutoff but does not verify minimal integer cases, maximal feasible cases, or adjacent boundary conditions.

Bottleneck in Domain-Specific Reasoning and Impractical Problem-Solving (23% of Errors). The model often demonstrates directionally plausible but insufficiently grounded reasoning: responses are structurally coherent and capture the general topic, but fail to satisfy the task's concrete requirements. A common pattern is that the model responds to the topic rather than to the question. As for legal tasks, those failures usually lie in inability to apply domain-specific legal norms or precedent-based reasoning. Models sometimes avoid hallucination by repeatedly emphasizing limited information, lack of access to external sources, or uncertainty, and fail to proceed with the conditional analysis expected by the rubric. This behavior suggests that surface-level coherence is not sufficient for domain-specific tasks.

Grounding and Factual Errors (17% of Errors). We frequently observe that factual grounding remains brittle including missing or incorrect task-critical anchors, such as policy dates, applicability conditions, statutory constraints, or technical boundary conditions. In many cases, the response is syntactically well-formed but lacks the specific evidence needed to justify the conclusion. This issue is particularly problematic in domains such as finance, law, and medicine, where small factual deviations can lead to a quite different reasoning result.

Instruction Violation (14% of Errors). The model may ignore output constraints, replace user-specified methods with more familiar templates, or shift from a requested neutral description to an opinionated recommendation. In some cases, the model may answer the substance of the question partially correctly while still failing this category because it ignored a required format, omitted mandated evidence, or did not follow the requested procedural approach. Such violations are especially harmful in benchmark settings that rely on strict rubric compliance. Overall, these failures indicate that improving domain-specific performance requires not only better general reasoning, but also stronger grounding, more robust verification, and tighter alignment with task-specific instructions.

Fact Hallucination and Refusal (10% of Errors). We also observe that the model either refuses to answer when information is incomplete or fills the gap with plausible but unsupported details. The former leads to overly conservative outputs that underutilize information available in the prompt; the latter introduces hallucinated facts, including numerical estimates or assumed future disclosures. Both behaviors reduce utility in settings where conditional reasoning is required. Ideally, the model should distinguish between missing information that invalidates a claim and missing information that still permits a constrained, conditional answer.

Calculation Errors (2% of Errors). The model can often perform isolated arithmetic steps, but struggles with comparisons, ratios, and relative impact assessment. As a result, it may report absolute changes without translating them into normalized quantities or effects, making it difficult to assess materiality. Relatively, the model may state that an effect is limited without providing the numerical comparison needed to support that claim. This limitation is especially salient in tasks requiring threshold analysis, sensitivity estimation, or comparison against a baseline.

5 Related Work

Agentic Benchmarks. Recent works like xbench target professional productivity tasks [6], while SWE-bench measures software engineering capability through GitHub issue resolution [11] and τ-bench evaluate conversational interactions under policy constraints [42, 3]. Vending-Bench and FutureX examine long-horizon coherence and future prediction in dynamic scenarios [2, 45]. Compared to $OneMillion-Bench, prior testbeds such as GDPVal [27], RLI [14], and APEX-Agents [39] target economically valuable tasks, measuring performance through proxies merely rather than directly quantifying the real-world earnings an agent is capable of delivering with reproducible evaluations. **Unlike GDPVal, RLI, and APEX-Agents, $OneMillion-Bench focus more on consistent delivery (§4.3), pareto optimality (§4.4), and diachronical analyses (§D.1) for realistic grounding.**

Reality-grounded Evaluations. A complementary direction evaluates agents in real-world environments where outcomes are determined by external reality. LiveTradeBench formalizes this paradigm using streaming financial markets, showing that leaderboard performance does not reliably translate into real trading outcomes [43]. Public competitions such as Alpha Arena further demonstrate the high variance and regime sensitivity of agent performance in dynamic markets [22]. Beyond finance, Lab-Bench evaluates agents in real scientific workflows involving experimental data and literature analysis [12]. While these assessments offer strong ecological validity, they sacrifice diagnostic resolution: failures are difficult to attribute to specific reasoning, retrieval, and tool execution issues.

6 Conclusion

In this work, we introduce $OneMillion-Bench, a testbed aimed at bridging the gap between static exam-style evaluations and the high-stakes demands of professional deployment. By integrating expert-curated workflows with a rubric-based assessment focusing on logical coherence, factual grounding, and professional compliance, $OneMillion-Bench sets a rigorous standard for evaluating the economic readiness of language agents. $OneMillion-Bench emphasizes the need to shift the evaluation focus from superficial correctness to a framework that prioritizes genuine decision-making capabilities for agent optimization.

References

[1] Anthropic. Introducing claude opus 4.6. https://www.anthropic.com/news/claude-opus-4-6, Feb. 2026. Accessed: 2026-05-07.

[2] A. Backlund and L. Petersson. Vending-bench: A benchmark for long-term coherence of autonomous agents. *arXiv preprint arXiv:2502.15840*, 2025.

[3] V. Barres, H. Dong, S. Ray, X. Si, and K. Narasimhan. τ^2-bench: Evaluating conversational agents in a dual-control environment. *arXiv preprint arXiv:2506.07982*, 2025.

[4] Biglaw Investor. Biglaw salary scale. https://www.biglawinvestor.com/biglaw-salary-scale/, 2024. Referencing Association of Corporate Counsel (ACC).

[5] ByteDance Seed Team. Seed2.0. https://seed.bytedance.com/en/seed2, 2026. Official model page. Accessed: 2026-05-07.

[6] K. Chen, Y. Ren, Y. Liu, X. Hu, H. Tian, T. Xie, F. Liu, H. Zhang, H. Liu, Y. Gong, et al. xbench: Tracking agents productivity scaling with profession-aligned real-world evaluations. *arXiv preprint arXiv:2506.13651*, 2025.

[7] S. Dou, M. Zhang, Z. Yin, C. Huang, Y. Shen, J. Wang, J. Chen, Y. Ni, J. Ye, C. Zhang, et al. Cl-bench: A benchmark for context learning. *arXiv preprint arXiv:2602.03587*, 2026.

[8] Google DeepMind. Gemini 3 pro. https://deepmind.google/models/gemini/pro/, Nov. 2025.

[9] inclusionAI. Ling-2.5-1t. https://huggingface.co/inclusionAI/Ling-2.5-1T, 2026. Hugging Face model card. Accessed: 2026-05-07.

[10] S. Jabbour, T. Chang, A. D. Antar, J. Peper, I. Jang, J. Liu, J.-W. Chung, S. He, M. Wellman, B. Goodman, et al. Evaluation framework for ai systems in" the wild". *arXiv preprint arXiv:2504.16778*, 2025.

[11] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. R. Narasimhan. Swe-bench: Can language models resolve real-world github issues? In *The Twelfth International Conference on Learning Representations*, 2024.

[12] J. M. Laurent, J. D. Janizek, M. Ruzo, M. M. Hinks, M. J. Hammerling, S. Narayanan, M. Ponnapati, A. D. White, and S. G. Rodriques. Lab-bench: Measuring capabilities of language models for biology research. *arXiv preprint arXiv:2407.10362*, 2024.

[13] A. Liu, A. Mei, B. Lin, B. Xue, B. Wang, B. Xu, B. Wu, B. Zhang, C. Lin, C. Dong, et al. Deepseek-v3. 2: Pushing the frontier of open large language models. *arXiv preprint arXiv:2512.02556*, 2025.

[14] M. Mazeika, A. Gatti, C. Menghini, U. M. Sehwag, S. Singhal, Y. Orlovskiy, S. Basart, M. Sharma, D. Peskoff, E. Lau, et al. Remote labor index: measuring ai automation of remote work. *arXiv preprint arXiv:2510.26787*, 2025.

[15] Medical Group Management Association. 2024 provider specialty rollups. https://mgmatraining.com/wp-content/uploads/2025/01/ProviderSpecialtyRollUps2024.pdf, 2024.

[16] K. J. Meimandi, G. Aránguiz-Dias, G. R. Kim, L. Saadeddin, A. Griffith, and M. J. Kochenderfer. The measurement imbalance in agentic ai evaluation undermines industry productivity claims. *arXiv preprint arXiv:2506.02064*, 2025.

[17] M. A. Merrill, A. G. Shaw, N. Carlini, B. Li, H. Raj, I. Bercovich, L. Shi, J. Y. Shin, T. Walshe, E. K. Buchanan, et al. Terminal-bench: Benchmarking agents on hard, realistic tasks in command line interfaces. *arXiv preprint arXiv:2601.11868*, 2026.

[18] G. Mialon, C. Fourrier, T. Wolf, Y. LeCun, and T. Scialom. Gaia: a benchmark for general ai assistants. In *The Twelfth International Conference on Learning Representations*, 2023.

[19] MiniMax. Minimax m2.5: Built for real-world productivity. https://www.minimax.io/news/minimax-m25, Feb. 2026. Official blog post. Accessed: 2026-05-07.

[20] MiniMax-AI. Minimax-m2.1: A sota model for real-world development & agents. https://github.com/MiniMax-AI/MiniMax-M2.1, 2025.

[21] Moonshot AI. Kimi k2.5: Visual agentic intelligence. https://www.kimi.com/blog/kimi-k2-5.html, Jan. 2026.

[22] Nof1. Nof1: Alpha arena. https://nof1.ai/, 2026. AI trading benchmark and research platform focused on financial markets. Accessed: 2026-05-07.

[23] OpenAI. Introducing deep research. https://openai.com/index/introducing-deep-research/, Feb. 2025.

[24] OpenAI. Introducing gpt-5.2. https://openai.com/index/introducing-gpt-5-2/, Dec. 2025.

[25] OpenAI. Introducing gpt-5.3-codex. https://openai.com/index/introducing-gpt-5-3-codex/, Feb. 2026. Official blog post. Accessed: 2026-05-07.

[26] OpenAI. Introducing gpt-5.4. https://openai.com/index/introducing-gpt-5-4/, Mar. 2026. Official blog post. Accessed: 2026-05-07.

[27] T. Patwardhan, R. Dias, E. Proehl, G. Kim, M. Wang, O. Watkins, S. P. Fishman, M. Aljubeh, P. Thacker, L. Fauconnet, N. S. Kim, S. Miserendino, G. Chabot, D. Li, P. Chao, M. Sharman, A. Barr, A. Glaese, and J. Tworek. GDPval: Evaluating AI model performance on real-world economically valuable tasks. In *The Fourteenth International Conference on Learning Representations*, 2026.

[28] K. Pearson. Vii. mathematical contributions to the theory of evolution.—iii. regression, heredity, and panmixia. *Philosophical Transactions of the Royal Society of London. Series A, containing papers of a mathematical or physical character*, pages 253–318, 1896.

[29] Perplexity. Sonar deep research. https://docs.perplexity.ai/docs/sonar/models/sonar-deep-research, 2025. Official documentation. Accessed: 2026-05-07.

[30] L. Phan, A. Gatti, N. Li, A. Khoja, R. Kim, R. Ren, J. Hausenloy, O. Zhang, M. Mazeika, D. Hendrycks, et al. A benchmark of expert-level academic questions to assess ai capabilities. *Nature*, 649:1139–1146, 2026.

[31] Qwen Team. Qwen3.5. https://qwen.ai/blog?id=qwen3.5, 2026. Official blog post. Accessed: 2026-05-07.

[32] D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, and S. R. Bowman. Gpqa: A graduate-level google-proof q&a benchmark. In *First Conference on Language Modeling*, 2024.

[33] R. Schwartz, R. Chowdhury, A. Kundu, H. Frase, M. Fadaee, T. David, G. Waters, A. Taik, M. Briggs, P. Hall, et al. Reality check: A new evaluation ecosystem is necessary to understand ai's real world effects. *arXiv preprint arXiv:2505.18893*, 2025.

[34] C. V. Snell, J. Lee, K. Xu, and A. Kumar. Scaling llm test-time compute optimally can be more effective than scaling parameters for reasoning. In *The Thirteenth International Conference on Learning Representations*, 2025.

[35] StepFun. Step 3.5 flash. https://static.stepfun.com/blog/step-3.5-flash, Feb. 2026. Official blog post. Accessed: 2026-05-07.

[36] Tencent Hunyuan Team. Tencent hunyuan 2.0 official release. https://mp.weixin.qq.com/s/wCJf5BOypPcKGxfp8jE9hg, Dec. 2025. Tencent AI Blog. Accessed: 2026-05-07.

[37] The Gemini Team. Gemini 3.1 pro: A smarter model for your most complex tasks. https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-3-1-pro/, Feb. 2026. Official blog post. Accessed: 2026-05-07.

[38] R. L. Thomas and D. Uminsky. Reliance on metrics is a fundamental challenge for ai. *Patterns*, 3(5), 2022.

[39] B. Vidgen, A. Mann, A. Fennelly, J. W. Stanly, L. Rothman, M. Burstein, J. Benchek, D. Ostrofsky, A. Ravichandran, D. Sur, et al. Apex-agents. *arXiv preprint arXiv:2601.14242*, 2026.

[40] xAI. Grok 4. https://x.ai/news/grok-4, July 2025.

[41] J. Xie, K. Zhang, J. Chen, T. Zhu, R. Lou, Y. Tian, Y. Xiao, and Y. Su. Travelplanner: a benchmark for real-world planning with language agents. In *Proceedings of the 41st International Conference on Machine Learning*, pages 54590–54613, 2024.

[42] S. Yao, N. Shinn, P. Razavi, and K. R. Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. In *The Thirteenth International Conference on Learning Representations*, 2025.

[43] H. Yu, F. Li, and J. You. Livetradebench: Seeking real-world alpha with large language models. *arXiv preprint arXiv:2511.03628*, 2025.

[44] A. Zeng, X. Lv, Z. Hou, Z. Du, Q. Zheng, B. Chen, D. Yin, C. Ge, C. Xie, C. Wang, et al. Glm-5: from vibe coding to agentic engineering. *arXiv preprint arXiv:2602.15763*, 2026.

[45] Z. Zeng, J. Liu, S. Chen, T. He, Y. Liao, Y. Tian, J. Wang, Z. Wang, Y. Yang, L. Yin, et al. Futurex: An advanced live benchmark for llm agents in future prediction. *arXiv preprint arXiv:2508.11987*, 2025.

Appendix

Table of Contents

A Authors **15**

A.1 Data Contributors 15

B Economic Value Audit of $OneMillion-Bench **15**

B.1 Calculation Method for Reproducibility 15

B.2 Task Economic Value List 16

C Category Distribution of $OneMillion-Bench **22**

D Further Analyses **24**

D.1 Diachronical Analysis 24

D.2 Correlation Analysis of Model Judge *vs*. Human Judge 26

D.3 Robustness to the cliff threshold 26

E Evaluation Details **26**

E.1 Prompt 26

E.2 Sampling Parameters 26

E.3 Evaluation Cost Estimation 27

F Case Examples **27**

F.1 Economics and Finance 27

F.2 Law 31

F.3 Healthcare and Medicine 34

F.4 Natural science 37

F.5 Industry 40

G Limitation **43**

G.1 Dynamic Expansion and Scenario Deepening 43

G.2 Automation of Fine-grained Process Evaluation 43

H Ethical Considerations **43**

A Authors

A.1 Data Contributors

We sincerely appreciate all experts for their invaluable data and feedback in advancing AI. Listed below are major contributors who agreed to be named (alphabetical by surname):

Mingyang Che, Jiarui Chen, Hanqi Li, Muzhen Liu, Hang Lu, Pan Wang, Shuo Wang, Wenzhou Wang, Chenyi Xie, Zhenjie Ye, Xia Yue, Jiayi Zhang, Yanru Zhang, Yongmou Zhao, Zishen Zhou.

B Economic Value Audit of $OneMillion-Bench

B.1 Calculation Method for Reproducibility

Formulation. The economic value of $OneMillion-Bench is estimated by summing the value of each question (*cf.* Eq. 4) from senior experts' required time cost and hourly wage listed in Table 10:

$$V_{\text{\$OneMillion-Bench}} = \sum_{q \in Q} V_q = \sum_{q \in Q} T_q^{\text{ExpertCost}} \times W_q^{\text{HourlyWage}}. \tag{4}$$

We provide a detailed table of task values (*cf.* Table 5) for calculating or reproducing the economic value results for any tested agent or model on $OneMillion-Bench.

Expert Cost. To ensure our valuation is grounded in reality, we gather time estimates for each task creation from 2 to 3 senior experts in their respective domains. This collaborative approach minimizes subjective bias, allowing us to derive a reliable mean estimation for task completion times.

Wage Anchoring. We developed a robust and multi-faceted wage reference that considers regional and sectoral nuances. For instance, in the United States, we utilized the latest Occupational Employment and Wage Statistics (OEWS) from the U.S. Bureau of Labor Statistics, complementing it with industry-specific reports for high-income roles, such as financial analysts[4] and medical specialists [15, 4]. In China, we tailored our estimates to reflect the economic realities of tier-1 cities, drawing data from recent labor market wage guidelines.

All calculations are standardized to an hourly basis using a typical work year of 2,080 hours (52 weeks $\times 5$ days $\times 8$ hours). We infer the implied hourly wage by dividing the reported annual salary by 2,080. To make compensation figures comparable across sources, we apply an overhead multiplier of 1.3 *only* when a source reports wages and salaries (*i.e.*, wage-only figures). This adjustment is motivated by the BLS *Employer Costs for Employee Compensation* (ECEC) release for June 2025 (released in 12, Sep, 2025): total employer compensation for civilian workers averaged $48.05 per hour worked, comprising $33.02 in wages and salaries and $15.03 in benefits. Benefits therefore account for about 31% of total compensation $\left(\frac{15.03}{48.05} \approx 31\%\right)$[5]. For sources that already report total compensation, we do not apply any multiplier.

Calculation for Reproduction. We provide the detailed list of economic value of each task in ($) USD or (¥) CNY unit depicted in Table 10 for reproducing the results in $OneMillion-Bench.

[4] https://www.biglawinvestor.com/biglaw-salary-scale
[5] https://www.bls.gov/news.release/ecec.nr0.html

B.2 Task Economic Value List

Table 5: Task economic value of $OneMillion-Bench

Task UUID	Domain	CNY	USD	Subset
f4cdab18-7636-4416-bcc0-9f40705687a5	Economics and Finance	5,793	–	cn
9709dc9b-ea0a-4e14-af06-3c636798b242	Economics and Finance	9,837	–	cn
50cea244-cd7b-4213-94f0-eb199e056eb5	Economics and Finance	6,664	–	cn
5e9d346b-e174-4f7c-bc0e-c0ddd3c0bb64	Economics and Finance	7,298	–	cn
0781ab70-3513-4da6-8032-a327aba51d26	Economics and Finance	3,862	–	cn
a4d2c2e8-9143-414a-86df-b76d565283c9	Economics and Finance	9,202	–	cn
c6fa3167-2fad-48c0-9160-62dde6424769	Economics and Finance	8,568	–	cn
4ab521dd-fcdb-4f61-b8e8-9eaf47fb392d	Economics and Finance	10,154	–	cn
3ef5a591-1158-42c7-8f49-427f7979634a	Economics and Finance	6,651	–	cn
b7e58dda-226a-42a8-8699-f98bbd5f60ad	Economics and Finance	6,346	–	cn
281a940f-16dd-4cff-bfd0-054c426e01bb	Economics and Finance	5,793	–	cn
4a961bc6-a0b7-4f2f-9a00-d5fc3365a19f	Economics and Finance	11,106	–	cn
1826f17c-0c59-44b9-925b-eeea8663429f	Economics and Finance	11,106	–	cn
5c786cc2-d1ed-48b3-8417-07c1910a5f94	Economics and Finance	6,222	–	cn
5a220c4d-1035-48aa-bc6f-f0f87cee79a8	Economics and Finance	4,720	–	cn
921ce736-5056-431e-bdab-769b161d3ccf	Economics and Finance	8,250	–	cn
b090fa25-d637-4f96-add1-546c49628b25	Economics and Finance	7,298	–	cn
472e9e99-f6c2-43df-b9e2-5e3feefaadda	Economics and Finance	5,077	–	cn
98da94c3-3733-4eca-b6c9-daf6929aa8c7	Economics and Finance	8,568	–	cn
3f500bde-303b-4b22-a11f-871eb1fbd705	Economics and Finance	9,520	–	cn
14e8f749-4d45-4fef-b9cf-c47788bfc908	Economics and Finance	8,885	–	cn
d08cefc6-dcb2-433b-88a4-e4ed3e9c19e1	Economics and Finance	6,346	–	cn
172516cf-4cff-49d3-90e4-30d8cbd285ca	Economics and Finance	3,218	–	cn
6a029f36-71d0-429b-a64c-9ef950d2686a	Economics and Finance	8,885	–	cn
16c841ac-f21b-4983-b48a-7248f3c8eb7b	Economics and Finance	4,760	–	cn
3100747e-ced0-4493-beaa-a37ee2b1ef97	Economics and Finance	6,007	–	cn
0d503359-64f3-47a8-9911-30baddd5eab7	Economics and Finance	4,291	–	cn
c1805f6c-c29e-41e8-a1c5-349bb14c113c	Economics and Finance	11,106	–	cn
61e05096-325a-4f3c-b1b4-74a34d0896d9	Economics and Finance	9,520	–	cn
ec9352fa-56c0-4a2f-b7aa-5cc2b3850f58	Economics and Finance	8,885	–	cn
034226e4-d37b-4cbf-b57d-74a018123f18	Economics and Finance	9,520	–	cn
39774cd5-aa36-428c-95fc-020ebc38ad8f	Economics and Finance	11,106	–	cn
21439c3b-52b4-43fd-a4ba-b5304051f9d0	Economics and Finance	5,712	–	cn
8cfaeaa6-f275-4e8e-bef8-8785855ed244	Economics and Finance	9,837	–	cn
9f8000ca-9a59-4f06-8107-0e223b65e826	Economics and Finance	10,789	–	cn
dabd5ce8-8348-4ea8-a697-3fb7376636af	Economics and Finance	3,218	–	cn
a3a7dbe4-5521-4c6b-8a42-c7ee61ded640	Economics and Finance	4,291	–	cn
bc8b6415-8fcd-4633-b7bb-764668230 36b	Economics and Finance	7,509	–	cn
11e6f0de-2dee-4a0a-bb5a-6211deb59bec	Economics and Finance	3,433	–	cn
559f2275-0ebb-4e53-8568-d9b6335bfdff	Economics and Finance	7,080	–	cn
e1bcc059-86a7-4b4d-8e5c-9de507d6ef1c	Economics and Finance	–	6,042	global
aaf8b644-94ea-4a68-8386-a44761be88a7	Economics and Finance	–	6,334	global
68795f96-1443-44d0-903c-020ca1d32c59	Economics and Finance	–	3,814	global
b342585e-9665-444b-9f42-4afd346b3955	Economics and Finance	–	4,486	global
bbdd25c4-f8cc-4208-bc4d-f9a9d56847d7	Economics and Finance	–	4,131	global
160179a5-f900-4c06-971b-b79229f44ad3	Economics and Finance	–	4,131	global
d4ed1bc5-eff1-4b41-b787-ec13b010a0cb	Economics and Finance	–	4,131	global
72539acb-922f-4e39-804f-20c2035d3c60	Economics and Finance	–	3,814	global
68f1fe22-77da-4d5b-886c-8fc4e18ad459	Economics and Finance	–	5,005	global
0f141139-c43b-486f-a9f9-3699547585d7	Economics and Finance	–	4,277	global
5e7c850f-2b48-437c-9dab-4e2e164d6373	Economics and Finance	–	3,679	global
e1b94c86-b6c9-43f6-8251-2e513e5efc52	Economics and Finance	–	3,840	global
13916e64-de22-4b8a-b790-62d3c1a19740	Economics and Finance	–	5,751	global
6946a41f-5b87-46fc-bf3a-83ffd2986b24	Economics and Finance	–	5,460	global
ebdcd944-aa53-4727-9591-3c8ce27bd000	Economics and Finance	–	8,117	global
38b7dfbd-df19-47a6-9e4e-f31e571d20b6	Economics and Finance	–	4,906	global
4d15d7a3-4f47-4c4d-a8db-a7c6fe0e75f7	Economics and Finance	–	4,771	global
2381e458-11f0-4878-a4c4-6442d3cde24a	Economics and Finance	–	5,309	global

Continued on next page...

Table 5 (continued)

UUID	Domain	CNY	USD	Subset
8077372b-e72d-43a3-bac4-3ff995040ba1	Economics and Finance	–	4,131	global
45435d84-9e68-40c9-b2f0-ce69b248c89b	Economics and Finance	–	3,259	global
0d8b4484-e80d-482b-8b4a-65803e60ecb3	Economics and Finance	–	5,606	global
06150229-a28a-4842-a93a-fa7dfeb9031d	Economics and Finance	–	4,423	global
db77b8bf-04c8-49bc-a8a4-578c41589636	Economics and Finance	–	3,948	global
f1dcbc6e-133f-4f1b-8b74-9e07f69d74f4	Economics and Finance	–	6,479	global
ccd2f927-5479-42b8-bb7d-07b16b7c5f8a	Economics and Finance	–	3,125	global
e8c50005-5621-46a4-8f1d-014317bac459	Economics and Finance	–	4,423	global
df72fe47-0a47-48d0-bf0c-fa39509908df	Economics and Finance	–	3,259	global
6c0bf4ed-e56a-4ca7-b21c-23038468133f	Economics and Finance	–	4,486	global
15b6abcd-80ab-42c9-ab11-aa03d2b216bc	Economics and Finance	–	5,040	global
8002472d-7ccd-4970-8e01-67e9a7cf793f	Economics and Finance	–	3,125	global
15ae0a17-0b9a-4f19-b57c-27b76cc1e8c8	Economics and Finance	–	3,814	global
67c5d15f-65d3-45ba-9df8-debcc08e0a47	Economics and Finance	–	3,410	global
972f6b74-3e6a-41eb-a7c8-b8e388999d8f	Economics and Finance	–	4,859	global
f0cfd1a0-efbf-43b9-826e-98ea95fee9d8	Economics and Finance	–	6,188	global
fbe57877-ec38-4cda-bd95-fb5c773662c4	Economics and Finance	–	4,351	global
5b070842-8ff7-484b-8ea7-98c3a19fe739	Economics and Finance	–	5,169	global
ea45e6d4-d05f-4372-9587-c77a38d6f250	Economics and Finance	–	4,568	global
7c5ca57b-b6f3-4b6b-85f3-73c60dd0878c	Economics and Finance	–	3,986	global
ad1240ff-6ede-4794-82af-0e2d31e487a9	Economics and Finance	–	4,131	global
d73d983f-e531-4bc9-8a17-9ace27450c09	Economics and Finance	–	3,948	global
14c80191-6712-46fb-a48e-0c3e4e714c7a	Law	6,180	–	cn
4a5571ee-41f4-4bab-aa0c-8220ae6b7191	Law	3,675	–	cn
98eb28f3-ec92-4a07-aafa-9997f96955aa	Law	4,343	–	cn
c902d3e0-c8a4-4610-90ec-b4e93da13ab9	Law	3,629	–	cn
4f44b9e3-06f8-4e04-8983-1cbec827fb7a	Law	4,510	–	cn
d3ea340a-8fcf-4cbf-9532-fd50e4ced0ac	Law	5,512	–	cn
9aea07f8-60c1-4ea4-bde3-d1562fc1aa1b	Law	3,508	–	cn
d4bf75d1-4e83-4f87-94f0-a60c44c8c16f	Law	5,679	–	cn
1b355e8f-25c3-44c5-8d76-55b791d686e4	Law	3,007	–	cn
a12269fe-0ae3-403f-8e4a-ec1278568307	Law	6,180	–	cn
a7a67d1c-d7fa-4e8e-9141-01f0b4649e01	Law	4,666	–	cn
86af7cd8-a582-4035-b0d2-788292786c90	Law	4,510	–	cn
4cca073f-5e11-40db-9a5f-8909144a2beb	Law	3,341	–	cn
746cfd85-18e9-4667-aa63-a61f88022f7f	Law	6,514	–	cn
bd130497-af77-4163-a49c-ac195d164685	Law	4,510	–	cn
5931cbd0-aa90-4e0d-8b37-2fac30b6c5a4	Law	6,347	–	cn
022812c3-d191-4ab5-82d7-cd4cbc8019f0	Law	5,512	–	cn
bf9888e6-f5ee-4d00-ab88-7209753d7d05	Law	5,011	–	cn
079fbcc9-3520-407e-83fa-623378b2f36c	Law	5,679	–	cn
dc8277a4-e3c9-45e8-a942-fe1f80273776	Law	3,174	–	cn
43a0f47c-21eb-4323-8eeb-a7da3e641758	Law	4,343	–	cn
4656b5dc-73de-405c-bccb-fcc77bf63cc1	Law	4,510	–	cn
fba1cdc6-e08d-440d-be8a-36f85a06fb70	Law	6,180	–	cn
5b7d77f0-4f75-4579-8338-3a9be4aeeaa9	Law	4,925	–	cn
907e27a2-8a94-4ac5-8686-7f2d839497f5	Law	3,675	–	cn
0397f35a-5e9e-4281-a74f-217f4df2786d	Law	4,343	–	cn
d0b5291f-ed30-4e32-aca6-b380f1748e35	Law	4,009	–	cn
64c29e74-abce-41b1-b275-22c055e5a6da	Law	5,846	–	cn
f63e490e-1205-4417-8325-d0b64811bbc6	Law	5,512	–	cn
0f0eac70-bb29-4f45-9423-7d8dc8fd4ce7	Law	6,514	–	cn
3ac266ca-c193-48fe-85c0-f4e9522961ce	Law	4,176	–	cn
dac34e8c-217c-447d-8694-6dbe34c0746c	Law	4,677	–	cn
7fc7f2ab-d56f-438a-b9ae-424323f96e0c	Law	5,011	–	cn
611387e2-7a44-4ee5-add7-2563a45b5264	Law	3,110	–	cn
46b5480b-5b9a-4d4c-b115-45bf60b616f0	Law	4,677	–	cn
448d98aa-660d-498a-8213-213f1065ff96	Law	5,178	–	cn
d7d9214f-ac7c-487d-852f-20845caa413a	Law	3,370	–	cn
dd459add-fe84-4de2-b177-f470389454a6	Law	3,842	–	cn

Continued on next page…

Table 5 (continued)

UUID	Domain	CNY	USD	Subset
0d3acaab-f7de-454b-bba9-fe4694acc77c	Law	5,011	–	cn
8124b83e-9ee9-4a99-b703-28bb492a3688	Law	4,510	–	cn
7da100c7-4054-419b-8226-c3e97dce0b54	Law	–	12,302	global
23d9a97d-aba9-4dd4-9a1f-35acde6c735f	Law	–	12,302	global
749e0b9b-eeed-4f57-a8ad-2759f4a268d9	Law	–	7,796	global
707c4831-8137-45a6-b889-3396695f8205	Law	–	9,239	global
d01ba3e0-4620-4baa-8b26-5d6597fbd7f0	Law	–	7,189	global
76fc9645-cb52-41f4-9971-21e67773b1f4	Law	–	5,543	global
3b763792-7c5c-42ac-81d7-b2b2d31309d3	Law	–	4,911	global
97dc696f-bb3b-4476-b818-f42a42c0f192	Law	–	7,391	global
61ee7292-5e3b-4bce-bd76-c97e0d4ffad2	Law	–	6,961	global
2d015f0d-4564-4a26-b5d4-fd2114fb3161	Law	–	7,189	global
926c1afb-e115-4e4d-831a-f857afc6a5c1	Law	–	5,431	global
92ecbb65-9f51-47f9-b918-d6ab932c70bc	Law	–	9,442	global
1bd5358d-f6ed-47cb-8a02-d303e2e1407d	Law	–	6,961	global
386cc233-cdf0-43cd-9614-c8e436e00ead	Law	–	7,999	global
d78a6473-ab98-4cb4-bdfe-c46ed5b67a08	Law	–	3,764	global
1070d2a9-1576-4e8a-aa25-d01d2f37e477	Law	–	12,302	global
dd3dca51-0b81-419c-ad75-c5c3ed7544d7	Law	–	8,201	global
7ad023af-e1fc-44e4-afc3-b04098ddb324	Law	–	5,341	global
42943063-a1b0-419b-a62f-0ee09c4d5785	Law	–	7,511	global
d728baca-5d5d-47fe-882d-dbc6e1de9f55	Law	–	7,999	global
68c0e9e4-4e01-4137-b62e-c523a3e0116e	Law	–	9,011	global
33e7a536-ac9f-465d-b34d-76b29f3512c0	Law	–	8,201	global
420150e9-93c1-4d5e-9760-966ebb66daf4	Law	–	8,404	global
dbbd68e9-9a20-42ff-ba8a-87218345f1cc	Law	–	6,961	global
136d58d2-bd74-4049-93fd-3df23cbdeb91	Law	–	9,442	global
6faea514-a56c-4aca-9414-593c34a1b7b1	Law	–	11,289	global
6183a0ce-401d-4763-a960-2509c2384765	Law	–	10,252	global
96c86547-d48b-4ca5-8303-8be965d4ebed	Law	–	7,999	global
a47b504a-b3f6-4e21-8fe4-30512c1db407	Law	–	3,627	global
ff42d583-5064-4cb3-8ecc-ab562d56c060	Law	–	6,758	global
588ca9fa-84b9-4ed8-962a-65b86e0500b5	Law	–	4,708	global
dc5967e8-18bb-4b4f-8c50-b0ce6eb2c87b	Law	–	9,442	global
bd0cfce5-9056-4697-a35e-bb6e13865a22	Law	–	5,543	global
e3b60852-b6ba-4eed-a2f0-e8ae2a364bc9	Law	–	7,796	global
6f9511f6-951a-4c97-ac10-694c0d201748	Law	–	4,506	global
02f114bf-c1ba-4d21-b8cc-c1c31ab495f9	Law	–	7,796	global
24804afe-da39-4fd7-908f-56e21ab03b9c	Law	–	12,302	global
580310a9-bb84-401f-96bc-55a1eaa5447c	Law	–	7,796	global
41af7f51-32e2-4a97-b4dc-fb2ea631ebdc	Law	–	3,902	global
330c4240-c6d4-4830-b1a1-788da3f69591	Law	–	5,138	global
443c4bff-5502-4658-91c4-0f5b26a4d4f0	Healthcare and Medicine	–	6,067	global
68922ab6-13c7-422c-9e0c-2429a4fafd7b	Healthcare and Medicine	–	7,660	global
ec20a6b0-7320-454e-8398-8f382a89efd2	Healthcare and Medicine	–	6,716	global
c437a033-7f62-473c-810d-35ed79d60df6	Healthcare and Medicine	–	11,622	global
a4119dbd-d5aa-4b1f-a215-d2f1b7285165	Healthcare and Medicine	–	7,320	global
3ecca542-c7a0-456d-8897-6c85f9732d11	Healthcare and Medicine	–	7,660	global
5cc4e65d-8984-4352-8dc4-f707c9da5eb1	Healthcare and Medicine	–	9,773	global
d9a3a37b-2fde-46f3-90c6-48a6d2215322	Healthcare and Medicine	–	10,376	global
df0b46a6-087d-4d4c-804e-dfbe7fc2f67f	Healthcare and Medicine	–	11,018	global
2b54a7d1-2d11-45ff-985e-5ce390ec0e1c	Healthcare and Medicine	–	9,169	global
669ccd28-509f-4e5b-a3c6-f4cc7d8930cf	Healthcare and Medicine	–	10,376	global
6a94f803-014f-4323-9c44-2bb80a59e19f	Healthcare and Medicine	–	4,658	global
c1c2c26f-0384-4f3c-82a3-5482b7009d56	Healthcare and Medicine	–	11,018	global
f003d40a-1190-409d-841e-84da045d24cc	Healthcare and Medicine	–	10,074	global
31fe657a-50a8-46c8-8186-f6b8da611584	Healthcare and Medicine	–	7,660	global
f223b123-621c-466f-b9fc-8e0cbe2785bd	Healthcare and Medicine	–	9,471	global
92ed8a85-f03b-458c-8b5a-9fea360ed649	Healthcare and Medicine	–	9,471	global
e1b922a9-ccd0-4a44-ae27-1d51c6cae4f2	Healthcare and Medicine	–	10,376	global

Continued on next page…

Table 5 (continued)

UUID	Domain	CNY	USD	Subset
d1aa95fa-4b01-4bde-a289-bb91bf489f79	Healthcare and Medicine	–	6,298	global
a6424187-9d45-4659-8a10-2f0f5e4524d4	Healthcare and Medicine	–	12,225	global
85a7c05e-2235-4d0c-ba8c-a2cca33ab401	Healthcare and Medicine	–	7,660	global
77b68a7d-c72f-4c21-88d6-36917feb5e85	Healthcare and Medicine	–	5,811	global
d8bd8164-b21f-4f6b-8ec6-7d058c514412	Healthcare and Medicine	–	7,660	global
0f464232-4413-4972-9e19-021d7d793161	Healthcare and Medicine	–	8,565	global
82664b2b-6471-4f66-a496-e35a67a18d40	Healthcare and Medicine	–	9,773	global
4b2c9234-229a-416b-9993-77b03b11d05e	Healthcare and Medicine	–	8,867	global
7bcf6aef-2235-45cd-8b02-02e8d9a00a39	Healthcare and Medicine	–	4,428	global
bb0a8654-2760-4fa0-9fa6-0e047c0d8d71	Healthcare and Medicine	–	11,923	global
4e906852-19f1-4380-9fc8-7f39af4ad3ca	Healthcare and Medicine	–	8,867	global
308de99f-a853-45a6-be11-67be5b113333	Healthcare and Medicine	–	8,397	global
26212219-9a4a-4afb-9cbe-819dbf0f73bf	Healthcare and Medicine	–	4,888	global
af865a52-51d3-4d3e-b170-d8d249e57f95	Healthcare and Medicine	–	8,263	global
e0921259-ac28-410d-94ad-6e795cd3250d	Healthcare and Medicine	–	4,888	global
5a2689ad-fbc9-45cd-b35f-2bdd5a33fb84	Healthcare and Medicine	–	6,716	global
3fc30be9-8aff-4c7e-afa3-d4c60c7b170b	Healthcare and Medicine	–	11,622	global
9fe8853f-42c5-41d4-8622-79e5b6a9a753	Healthcare and Medicine	–	8,263	global
d19a66bd-6552-4d0f-9574-6128f4f7ff05	Healthcare and Medicine	–	5,119	global
96d762f1-aaee-473d-a092-1024ce63872e	Healthcare and Medicine	–	6,067	global
01995bb2-e019-4195-844e-4012250583c9	Healthcare and Medicine	–	6,113	global
7db19fe4-4164-45d0-947e-61bef3496e75	Healthcare and Medicine	–	4,658	global
673490a8-a490-4809-8a7c-f2950685c331	Healthcare and Medicine	3,050	–	cn
86b11c02-b600-44ff-9806-176a723cdddf	Healthcare and Medicine	4,051	–	cn
f6d9a7e6-6d20-4f89-9937-1c9ba7836e73	Healthcare and Medicine	3,552	–	cn
d99d37b8-14d6-4df3-ada7-f5b3721747cb	Healthcare and Medicine	6,146	–	cn
4c394bc9-4f72-4b9d-b5ec-7b26968c8d68	Healthcare and Medicine	3,871	–	cn
7b7fe40e-e5d7-4cc0-887b-3c1d216bee06	Healthcare and Medicine	4,051	–	cn
2008063e-3dd5-4b76-831f-f06ef4e7d739	Healthcare and Medicine	5,168	–	cn
611c260e-93ef-4391-bbdd-0beda5f267de	Healthcare and Medicine	5,487	–	cn
c6b234bb-2142-4b26-993d-a0c2984a3943	Healthcare and Medicine	5,826	–	cn
dcfde4cd-fcb9-4390-94ea-3b8016158250	Healthcare and Medicine	4,849	–	cn
7cc331b2-297b-4311-ba03-a5ed0a27996b	Healthcare and Medicine	5,487	–	cn
c91c56f9-e432-484a-aaeb-867c7143d4bd	Healthcare and Medicine	2,342	–	cn
908f40e1-2d85-4e57-9320-cfc6fc450037	Healthcare and Medicine	5,826	–	cn
eb90ce6e-8a53-4fc4-8b71-4fc2f6913cdc	Healthcare and Medicine	5,328	–	cn
ce8ed8a0-8312-4464-ac08-cd33095eb019	Healthcare and Medicine	4,051	–	cn
81bf36b9-d2b6-40e6-9f4e-870fa0212b31	Healthcare and Medicine	5,008	–	cn
44847f8f-e10a-493b-a867-e505808f510f	Healthcare and Medicine	5,008	–	cn
b6502878-bfac-4be7-9851-d7364dd2ed3c	Healthcare and Medicine	5,487	–	cn
cd123b23-75a5-4619-8b9e-c77c7f5d9672	Healthcare and Medicine	3,165	–	cn
9a22b80c-9d00-488b-9e1e-7f9579efe2c2	Healthcare and Medicine	6,465	–	cn
12ec26fa-84c0-4639-b5c5-9e00c62dca0c	Healthcare and Medicine	4,051	–	cn
06be9028-f633-4737-98f3-b523421b1276	Healthcare and Medicine	3,073	–	cn
1135b8c9-60eb-4cd1-abfa-3d11bd522bb8	Healthcare and Medicine	4,051	–	cn
653c82d5-3dfa-44f7-b570-ca9ae59aa715	Healthcare and Medicine	4,529	–	cn
3107f221-d3cc-4401-b0e1-0b2fe36264f4	Healthcare and Medicine	5,168	–	cn
e71191ec-40f7-4156-b68d-b70754672fc4	Healthcare and Medicine	4,689	–	cn
ba4bb671-0fcc-4f6d-bb37-ea12aab8e5d8	Healthcare and Medicine	2,226	–	cn
fe491b1f-b37b-4f37-8e93-509ebcd000e9	Healthcare and Medicine	6,305	–	cn
8eb9fe93-8198-491c-a0ae-2d59b62b635a	Healthcare and Medicine	4,689	–	cn
f0597e81-d2ba-4069-a6de-8f86d29964e4	Healthcare and Medicine	5,826	–	cn
81e7a86e-6dfc-4a5b-8fdc-281924e24a45	Healthcare and Medicine	3,392	–	cn
b625b3c1-a92d-4234-911a-dfab5072dd51	Healthcare and Medicine	4,370	–	cn
1d156a89-d475-4c7a-8f19-ffc4a8514323	Healthcare and Medicine	2,457	–	cn
b9616065-9034-471b-8410-d53f231f1b5a	Healthcare and Medicine	3,552	–	cn
4bb63011-2170-4483-b171-2c74d2bf6102	Healthcare and Medicine	6,146	–	cn
dd646f83-7a44-4e24-8240-f626624fa179	Healthcare and Medicine	4,370	–	cn
d97af248-9a73-4a5b-8e9b-a406004cb5b2	Healthcare and Medicine	2,573	–	cn
578de37e-4cff-4436-8c44-d2cdc176d598	Healthcare and Medicine	3,050	–	cn

Continued on next page…

Table 5 (continued)

UUID	Domain	CNY	USD	Subset
de587f91-b1b2-4ec5-aa58-e48b2c9e5fb9	Healthcare and Medicine	3,232	–	cn
8ab658c4-f8c5-4533-9025-1dbcc2dad0ae	Healthcare and Medicine	3,232	–	cn
865cca91-9fca-4611-9dfe-2de5eec3ca41	Natural Sciences	–	1,340	global
fe5662f9-a6d6-4aad-9449-5156e61b9cec	Natural Sciences	–	2,147	global
944d060e-e3af-4f44-9ae4-51920e6a7898	Natural Sciences	–	2,971	global
68bfa996-706b-43e1-8a91-1af695ba5bb5	Natural Sciences	–	1,340	global
659a8308-83be-434b-9a74-d0d60e72b4d1	Natural Sciences	–	2,391	global
f5bd1ec1-b1bf-49b2-be47-c2c5ecc5d1df	Natural Sciences	–	1,730	global
32809395-def0-4e6b-afa2-b80a37620384	Natural Sciences	–	3,378	global
eae3cc5a-49e4-4a07-bc7d-e01ada5a0847	Natural Sciences	–	2,042	global
47bd8a68-2c91-4ceb-af3c-7f113bf4d213	Natural Sciences	–	2,066	global
c300a2f7-7b4c-4cf7-8385-6f49834c8a09	Natural Sciences	–	1,340	global
97a49948-4c98-41ed-9092-1ef88bc09e1f	Natural Sciences	–	1,214	global
228e062e-2dfb-450a-828d-446f5c18bb9a	Natural Sciences	–	1,277	global
3452218f-1d0d-45fc-ac72-a0f0d6fa17e5	Natural Sciences	–	1,853	global
a7024d80-956d-491c-b1bb-8567c7571999	Natural Sciences	–	1,466	global
8ca39d59-ba11-48de-8c7a-27a5d4f7935a	Natural Sciences	–	1,600	global
431c7ede-5c91-4a49-b4b3-cc94229822fc	Natural Sciences	–	2,428	global
a6f3f009-2786-4347-8b13-fa7e614695ab	Natural Sciences	–	1,979	global
2daed9dc-b3fa-4724-b0da-470da4e672a7	Natural Sciences	–	1,403	global
bea6d62b-80ec-4d95-81e8-c3f5952d63d2	Natural Sciences	–	1,853	global
31777574-0def-4e66-ad92-4274013cca63	Natural Sciences	–	1,151	global
833a9fb2-537a-42cd-ad0e-86b0df8ca415	Natural Sciences	–	1,916	global
274f5e8a-056f-4585-b0fb-2564868b793a	Natural Sciences	–	2,554	global
2e32c0a9-9382-4943-8594-309d29187539	Natural Sciences	–	1,214	global
b2d97709-e4f5-4bba-aee5-6676ca0ecdc2	Natural Sciences	–	1,403	global
b24a5d38-b539-4653-9cdc-dc3ca4e1f151	Natural Sciences	–	1,916	global
6bfba637-bd4f-469a-9bc5-6b92a55a3e59	Natural Sciences	–	1,600	global
bf61d44d-97fe-4365-9fed-8d3ac8440f54	Natural Sciences	–	1,151	global
585708d8-25c5-4e91-a779-5cf6e6fb1bf2	Natural Sciences	–	2,744	global
5fabbd4d-2295-4426-87c4-7fb735f4b657	Natural Sciences	–	2,635	global
404b288d-d5d8-4568-8de4-1f5ecedba486	Natural Sciences	–	1,403	global
2929294b-54cd-4fe1-8d28-2ae6f7c81e03	Natural Sciences	–	1,466	global
923f1733-de66-4d8f-a9c3-603a8b4b7158	Natural Sciences	–	1,974	global
8f8101f1-f874-4058-a523-5f01bbbec0be	Natural Sciences	–	1,663	global
3227a346-1115-4c4d-a0b6-bc5cd3ea914f	Natural Sciences	–	2,491	global
3cbe5faf-28b2-457c-bcd7-af179f5aec83	Natural Sciences	–	2,491	global
6e9131aa-dfd5-41e5-a2cf-d01041e895f5	Natural Sciences	–	1,340	global
dc4db810-7b0e-4110-bfb1-d53471680291	Natural Sciences	–	1,600	global
e92ecabf-a10a-46e1-98b2-a15d1ae073ac	Natural Sciences	–	3,516	global
dd20f5a3-670b-4fef-96f4-15d5a4186492	Natural Sciences	–	2,890	global
d4f2f740-c9b8-46f5-859d-b9e0fdb84f79	Natural Sciences	–	1,340	global
fd46a64b-90d3-481f-86da-8820d6cc7a38	Natural Sciences	1,751	–	cn
644d838a-bfeb-4043-b595-748527157415	Natural Sciences	3,485	–	cn
3e2cc4d7-0a4b-4862-bcf8-905535bf1829	Natural Sciences	4,823	–	cn
8e8151b8-1e45-411f-ab02-41614bb7400e	Natural Sciences	1,751	–	cn
6ca999e2-cdd0-4691-89d1-7534f5b564ce	Natural Sciences	3,882	–	cn
d2b65bf0-3872-4d24-8e52-1dc386fca06d	Natural Sciences	2,808	–	cn
dac5282c-448c-44f3-9f76-48e823852d17	Natural Sciences	5,484	–	cn
ee952885-9c32-4085-ac10-aa1536ade2ef	Natural Sciences	2,668	–	cn
d042f45e-f76a-4e39-ad94-bfd68a67e578	Natural Sciences	3,353	–	cn
2f5d1c57-c071-4b73-a226-d146d8b59c68	Natural Sciences	1,751	–	cn
b4d5950c-18f3-42f6-8f9b-6867c9949136	Natural Sciences	1,586	–	cn
059e1e9e-6e3e-4518-b880-b59173f1480e	Natural Sciences	1,669	–	cn
438aaf2a-eab2-40eb-b3d6-1ca4368585b8	Natural Sciences	2,420	–	cn
2c32b379-8c60-4648-97e5-c897dc46a623	Natural Sciences	1,916	–	cn
5ce6255b-3526-4916-9681-fcb5d6810da0	Natural Sciences	2,091	–	cn
6591d711-404f-48d0-ac7f-10f7094caeaf	Natural Sciences	3,172	–	cn
546dc745-d460-41a1-a9a0-0ecc446a31d6	Natural Sciences	2,585	–	cn
cd276d89-416c-4490-b688-3c79424c9699	Natural Sciences	1,833	–	cn

Continued on next page…

Table 5 (continued)

UUID	Domain	CNY	USD	Subset
76ea552b-e0f8-46f8-9203-e75903181843	Natural Sciences	2,420	–	cn
611d8974-aed7-45a1-b201-b1d2b380f82e	Natural Sciences	1,504	–	cn
3604da43-a774-4bd1-ae74-9d99a72354b1	Natural Sciences	2,503	–	cn
810bf406-5b65-4f47-b3f4-c62e507a08ee	Natural Sciences	4,146	–	cn
c3c93440-64ac-401e-8aa3-a0027dce2b6a	Natural Sciences	1,586	–	cn
07167000-3548-4fd7-ac80-0e12d54ac72f	Natural Sciences	1,833	–	cn
e73d9f41-3ccf-45e6-a387-e00472bfedd7	Natural Sciences	2,503	–	cn
d78cd986-1823-40b4-bf9d-931726a5d526	Natural Sciences	2,091	–	cn
938165db-06c0-428b-bdda-1f14ac25dbff	Natural Sciences	1,504	–	cn
44f67f0a-d62b-4ff0-b985-4a45d84e8911	Natural Sciences	3,584	–	cn
1b32355f-cd10-4e1f-aa8e-28cf4b5f2848	Natural Sciences	4,278	–	cn
949ac7a6-6b7e-46a1-8af1-c2963432b236	Natural Sciences	1,833	–	cn
66bf0a68-2d9b-457c-8817-19e13401cd2d	Natural Sciences	1,916	–	cn
280f0df8-7242-426a-91ac-5d61f1b48832	Natural Sciences	3,204	–	cn
d1c4b15f-9721-4827-840c-c6510a32b8dd	Natural Sciences	2,173	–	cn
2bc64a8e-1210-4dfd-aa30-ba02eceedd5a	Natural Sciences	3,255	–	cn
8622c778-80b1-442c-8997-de6e2405dd59	Natural Sciences	3,255	–	cn
1fc999bd-c4e5-4d04-b638-bfbb866d508f	Natural Sciences	1,751	–	cn
6de37f4f-8e50-4b1e-b5d2-8dbbcea8c3ca	Natural Sciences	2,091	–	cn
1dba23ad-c86b-4e82-a0e0-fa6c3d649538	Natural Sciences	4,594	–	cn
54e0d81e-67ec-4370-b886-9864ba328add	Natural Sciences	4,691	–	cn
457f5ca4-ba92-42b2-a82e-c4e6678e4f04	Natural Sciences	1,751	–	cn
9c2672f2-430f-44cc-823b-2784f3a84a2b	Industry	–	5,285	global
36d1050a-72a1-4210-94c8-cd654bc25132	Industry	–	2,469	global
44b07f0a-d6c2-4b74-bfdd-2c7b5a87e1c7	Industry	–	3,645	global
54a18a27-264b-42c7-9762-c030dc97bfe1	Industry	–	3,165	global
56cc66ac-2bde-4ce0-9c9e-8c816026020c	Industry	–	2,469	global
f0bc2660-fdb2-4e05-9766-e8db285d52e2	Industry	–	1,458	global
64b0f439-36d8-4fa1-8e81-81d37447cccf	Industry	–	3,350	global
6c34d48e-cc4e-4193-be63-a1094f541fe8	Industry	–	3,969	global
bba510cb-38c5-4c70-88e3-14780a0491c6	Industry	–	4,860	global
ee809a60-d4e2-4514-a3b5-175eb591f21e	Industry	–	4,125	global
f0bfdae8-df07-4c23-8ac0-08abcd0329be	Industry	–	4,108	global
89055e11-d0a5-4e7b-8d15-e1002f4cae54	Industry	–	2,259	global
302703db-b1c9-43e1-9684-c524139d6545	Industry	–	1,588	global
fad912f4-b576-4cd0-b659-9530ee97ade2	Industry	–	2,382	global
2b4583f1-f332-43c6-999e-9166e6d2c6e5	Industry	–	4,860	global
591d5f51-ed05-4c50-8abd-7a79c1333497	Industry	–	2,670	global
58a83321-05c2-4d66-86d0-1fbca0dc4741	Industry	–	2,382	global
d6357459-83e7-4bb6-8cbd-d2ff97c46a2f	Industry	–	2,110	global
0d785b81-fc88-438c-964b-e4c8256f9f42	Industry	–	1,827	global
550f71ad-94fe-40a0-9688-6acad415e987	Industry	–	3,089	global
5e785915-8448-461e-b663-9a8c9c777c4b	Industry	–	1,746	global
e2487a94-a128-4b67-9ed9-3fb056e28d5b	Industry	–	3,263	global
6b4ea30a-e9f1-4a30-a3bc-c6c006354ade	Industry	–	4,850	global
d46c9edd-b2e7-41f0-83a9-faf219653650	Industry	–	3,089	global
6267d153-72b8-45ef-81a5-d88db387096c	Industry	–	2,295	global
8276aec4-036f-459b-8770-2f15f693edd2	Industry	–	4,260	global
54246069-1446-41a1-8d17-89a0e5c52dc0	Industry	–	1,849	global
0defb49c-e4f5-451b-9821-807d77e8c8b3	Industry	–	1,530	global
025bf5b1-99d0-4d08-872e-5c4c7a77c5dd	Industry	–	4,500	global
c68fc03a-3fdc-4cca-824f-f14db3730027	Industry	–	2,475	global
1fb6bbf5-394d-49c7-ae64-bc374b0fcc3b	Industry	–	1,849	global
dcc3dbb7-0cbb-4f75-ae06-0efbb8987ad0	Industry	–	1,827	global
71f76252-1db0-41f8-93bf-338a89f17341	Industry	–	1,827	global
d6344516-e4f6-42b3-a2a6-f1329ed16dbf	Industry	–	1,458	global
364cfadc-3764-4ffd-a2c2-2688ae800950	Industry	–	3,645	global
1dbd3e38-1c81-4cf1-94eb-d1d24f9a2d49	Industry	–	2,382	global
6129a42f-a16c-4a28-8912-c60332e70406	Industry	–	2,295	global
deabcd1e-64ae-4659-82ac-646b517cdfc2	Industry	–	2,951	global

Continued on next page…

Table 5 (continued)

UUID	Domain	CNY	USD	Subset
b0cc7823-86a6-4a3e-acdd-0d20af89806e	Industry	–	2,106	global
fd1f36d1-c29a-4077-91f4-b0d23cb38f5b	Industry	–	1,899	global
ebb94405-b0af-4485-8efa-c95c3971a7b1	Industry	9,020	–	cn
42f2f642-deb1-46c6-a47e-b78c51b2cbcc	Industry	4,213	–	cn
a961442b-491d-469e-80e6-37c6b0301563	Industry	4,401	–	cn
c2a430c0-7f98-4ab5-af84-93824c790d72	Industry	3,821	–	cn
3aae3e0f-2c68-4575-86e4-015a146bc88e	Industry	4,213	–	cn
7c72e1b6-0eab-4e02-9a67-0661576dd212	Industry	1,242	–	cn
2ddd40b6-f110-47f4-ae3d-3fee5b01716f	Industry	5,716	–	cn
da515c84-7adc-4290-9807-67c26de850d5	Industry	6,774	–	cn
b5f5aecd-7ae0-4e6e-97ec-8dce1101d9bf	Industry	5,868	–	cn
520cdb68-0103-4ee3-80b5-0466a392a408	Industry	4,980	–	cn
ea67f32d-c827-438f-951e-17206a6f196c	Industry	3,637	–	cn
b29b17f5-d321-494e-a7fa-c4ec4fdb9a4f	Industry	1,924	–	cn
226b82d8-e406-4f1d-bd92-cb0adfb5374f	Industry	2,710	–	cn
088a6376-95c1-4b3b-a45e-4a5d19ea4b6e	Industry	4,064	–	cn
ab801e6d-7974-4cc6-b63a-36ba0faa0575	Industry	5,868	–	cn
d29e29db-affb-494c-b5b7-1b193cff24bd	Industry	3,224	–	cn
94e8ae5c-1bf8-49bc-bd40-f774644cc7f1	Industry	4,064	–	cn
7e0ab98e-379e-4b1e-acff-3ff37bffaf32	Industry	3,600	–	cn
6fd07669-8d90-4ace-81ea-9a1ec49cf9aa	Industry	1,556	–	cn
39f36658-5239-4536-a9cd-d0270b652111	Industry	5,271	–	cn
38b981dc-4f80-40ee-b568-0ed6bf129f62	Industry	1,487	–	cn
b7f472d0-1733-45f6-86f2-04d050be121c	Industry	5,568	–	cn
c4c00b09-7166-4d8c-b03d-273a9b43da6f	Industry	8,277	–	cn
84ec9ed0-abae-45d9-a303-935b47f23954	Industry	5,271	–	cn
467db57d-6c16-468f-b97b-cc33c94891dd	Industry	3,916	–	cn
b0d00294-81d8-453f-b8e1-76543c490717	Industry	5,143	–	cn
e79e0354-4848-4896-8c61-47ae21566204	Industry	3,155	–	cn
d6b99b4e-c7f9-4cab-8064-e782ad87115f	Industry	1,303	–	cn
3c8f83a2-96db-42b2-812d-643cfb1d99e4	Industry	5,433	–	cn
cce48b66-20eb-4d4a-9dfb-fff4e75968f8	Industry	2,108	–	cn
c037de9f-eb16-452e-832d-0af01f0ff50e	Industry	3,155	–	cn
4bd30355-9179-4db2-b19f-acfe08b71833	Industry	1,556	–	cn
a708701f-f2c5-40a6-b14d-d2f77f8924c9	Industry	1,556	–	cn
e771ab18-f120-4c54-8450-aa2367b71bf6	Industry	1,242	–	cn
c545314e-f070-43c7-bec7-c426687d27cf	Industry	4,401	–	cn
9a88315b-27e8-4263-a73a-4016befe01a4	Industry	4,064	–	cn
af4dcf04-29fe-4ba3-8d2f-d1bdae0406ab	Industry	3,916	–	cn
eb7638ac-035f-4b6c-a207-ebc6804aca3a	Industry	2,613	–	cn
34b542fb-9418-4485-a235-67c9f6eb505f	Industry	1,865	–	cn
bea3bcd5-4ef3-47f2-b563-7300ff506a2a	Industry	1,617	–	cn

C Category Distribution of $OneMillion-Bench

We identified a total of **92** unique third-level subdomain tags in $OneMillion-Bench in five high-level domains. The non-synonymous tags across languages are kept as separate entries

Healthcare and Medicine (25 tags):

- Cell and Gene Therapy (CGT)
- Gastrointestinal Surgery
- Pathophysiology
- Hepatobiliary and Pancreatic Surgery
- Cardiothoracic Surgery
- Thyroid and Breast Surgery
- Pathology

- Obstetrics and Gynecology
- Rheumatology and Immunology
- Biopharmaceuticals (Macromolecules)
- Orthopedics
- Molecular Biology
- Ophthalmology
- Medical Genetics
- Dentistry
- Emergency and Critical Care Medicine
- Cardiovascular Medicine
- Neurology
- Medical Laboratory Science
- Respiratory Medicine
- Immunology
- Endocrinology and Metabolic Diseases
- Oncology
- Nephrology
- Neurosurgery

Industry (12 tags):

- Semiconductors
- Telecommunications
- Machine Learning
- Backend Development
- Databases and Data Engineering
- Civil Engineering
- Chemical Engineering and Materials
- Architectural Design
- Systems / Embedded / 3D Rendering
- Frontend Development
- Machinery and Automation
- Materials

Law (19 tags):

- Criminal Defense
- Contract Disputes
- Restructuring / Financing / M&A
- Corporate Governance
- Data Compliance/Cybersecurity
- Financial Regulation
- Government Regulation
- Copyright
- Law-Other
- Trademark
- Patent
- Labor Law
- Marriage/Inheritance

- Criminal Compliance
- Securities & Listing (IPO)
- Funds & Asset Management
- Public International Law
- Private International Law
- Tort Disputes

Economics and Finance (18 tags):

- Equities
- Bonds
- Mergers & Acquisitions
- Management/Consulting/Business Analysis
- Risk Management
- VC/PE
- Accounting
- Investing-Other
- Equity Financing
- Commodities
- Consumer Finance
- Corporate Banking
- Other
- Cryptocurrency
- Derivatives
- Macroeconomics
- Life Insurance
- Quantitative

Natural Science (18 tags):

- Condensed Matter Physics
- Physics-Other
- Organic Chemistry
- Inorganic Chemistry
- Materials Chemistry
- Physical and Theoretical Chemistry
- Classical Physics
- Quantum Physics
- Applied Mathematics
- Mathematics-Other
- Microbiology
- Molecular Biology
- Molecular and Cell Biology
- Cell Biology
- Biochemistry
- Genetics and Genomics
- Ecology
- Other

D Further Analyses

D.1 Diachronical Analysis

Economics & Finance

Background

You are an actuary, currently working in the field of reserve valuation, specifically focusing on the calculation of various statutory reserves and reserves for financial reporting. Currently, due to the transition to IFRS 17, the calculation methods for various reserves have undergone significant changes. Your supervisor wants you to conduct the following research on the calculation of individual life insurance reserves and suggests that you refer to the contents of the fifth edition of the book 'Statutory Valuation of ILA Contracts'.

Question

1. Please list the four methods used by life insurance companies to calculate reserves prior to the advent of the Valuation Manual, and provide a detailed background on the characteristics of these calculation methods regarding assumptions such as premiums and operating expenses.
2. Please indicate which of the aforementioned methods utilize the concept of Expense Allowance. Based on what practical operational characteristics of insurance companies did the concept of Expense Allowance arise? What factors are typically considered in its subsequent amortization adjustment?
3. In practical work, it is often necessary to estimate reserve valuations between years. Using a term life policy as an example, please explain how reserves should be estimated under the assumption of continuous dates, identifying the specific method typically used for this estimation.

Rubrics

Factual Information

+9 List the four reserve calculation methods prior to the advent of the Valuation Manual: Net Level Premium Method (NLP), Full Preliminary Term Method (FPT), Modified Reserve Method, and Commissioners Reserve Valuation Method (CRVM).

+7 Elucidate that the Full Preliminary Term Method (FPT) sets the first-year net premium as the 'net premium for one-year term insurance,' implying that the first year covers only the cost of risk for that year.

+5 Point out that expenses (and lapses) are not explicitly assumed in the Net Level Premium Method (NLP) valuation formula and NLP is calculated solely based on expected future business cash flows.

Analytical Reasoning

+8 Analyze the reason for the proposal of the Expense Allowance concept, namely that the first-year acquisition costs of life insurance contracts are extremely high, necessitating a mechanism for offsetting and amortization.

-4 The model failed to point out that the Force of Interest method is typically used for estimation.

Structure and Formatting

-3 The formulas provided do not use Markdown formula blocks or LaTeX syntax, but instead use plain text or garbled characters, resulting in mathematical symbols that cannot be correctly read.

Law

Background

The UK government, through its intelligence agencies, has ascertained that a cohort of migrants from Northern Nigeria harbors extremist religious views, posing a threat to British society. Intelligence sources further revealed a conspiracy by said migrants to perpetrate terror attacks across the United Kingdom. Acting swiftly, the British Prime Minister persuaded Parliament to establish a Tribunal of Inquiry to investigate the matter. Furthermore, the Prime Minister contacted the President of Nigeria, advising the establishment of a National Tribunal of Inquiry to investigate religious extremism within that country.

Question

Discuss the legal basis for the President of Nigeria to establish a National Tribunal of Inquiry to investigate religious extremism in Nigeria.

Rubrics

Factual Information

+10 Cite the provisions regarding executive powers pursuant to Section 5 of the 1999 Constitution of Nigeria.

+9 Point out that the powers of the President of Nigeria are circumscribed by the legislative scope of the National Assembly.

+8 Point out that the Tribunals of Inquiry Act is deemed an existing law pursuant to Section 315 of the Constitution, yet it lacks constitutional validity at the federal level.

-3 The response includes suggestions regarding administrative engagement by security agencies, which constitutes an irrelevant and extraneous point.

Analytical Reasoning

+6 Argue that the National Assembly lacks the jurisdiction to enact legislation regarding Tribunals of Inquiry applicable to the entire Federation.

-2 The model incorrectly discusses the legal basis from the perspective of international or diplomatic cooperation, rather than strictly based on Nigerian domestic law.

Structure and Formatting

-4 The model fails to point out that the only remedy for establishing a National "Tribunal of Inquiry" is a constitutional amendment, or fails to mention the futility of the National Assembly enacting special laws.

Healthcare & Medicine

Background

Hypothesis: Do specifically enriched bacteria in the gut of Crohn's disease (CD) patients inhibit the expression of the bile acid transporter ASBT via the activation of inflammatory signaling pathways, thereby reducing intestinal bile acid reabsorption and causing luminal bile acid accumulation? Does this subsequently promote Clostridioides difficile spore germination, ultimately exacerbating the incidence and recurrence of CDI?

Question

Please design an experimental protocol to verify this mechanism.

Rubrics

Factual Information

+8 Explicitly propose causal perturbation strategies capable of directly testing the necessity of key nodes, and design clearly controlled functional validation experiments based on these. Specifically, the design must employ loss-of-function approaches targeting ASBT or its upstream inflammatory signaling nodes (e.g., inhibitors or KO mouse models) to compare changes in bile acid transport function or relevant phenotypes pre- and post-perturbation, thereby determining the causal status of the molecule in the hypothesized mechanism. Mere observational detection or correlation analysis does not constitute valid causal verification."

+7 Utilize Inflammatory Bowel Disease models (DSS or TNBS) for mechanistic exploration in animal experiments.

+6 Explicitly define clinical cohort grouping and follow-up design. The model must strictly categorize CD patients into 'CD+CDI' and 'CD' groups, and conduct prospective follow-up on the CDI group to distinguish between recurrent (rCDI) and non-recurrent subgroups.

Analytical Reasoning

+8 The experimental design establishes clear control groups in key mechanistic validation steps to distinguish 'effects mediated by target bacteria or inflammatory signals' from background effects; for example, including non-colonized controls, non-target bacteria controls, or non-induced inflammation controls, ensuring that changes in ASBT expression and relevant phenotypes are attributable.

-6 The experimental design fails to detect key molecular indicators (e.g., ASBT protein/mRNA expression, intestinal bile acid concentration, inflammatory cytokines), or relies solely on macroscopic phenotypic observations (e.g., recurrence rate) without molecular-level validation.

Structure and Formatting

-7 The experimental protocol lacks verification of intermediate mechanistic steps (e.g., omitting detection of ASBT expression, bile acid reabsorption, or spore germination), directly linking microbiota changes to CDI recurrence (functional outcome).

-9 Substitutes specific mechanistic action with macroscopic disease or ecological background descriptions.

Figure 7: Sample cases of different domains with varying rubric weights and tags.

As depicted in Figure 8, all models consistently achieve higher performance on temporally agnostic questions than on weakly sensitive and sensitive ones, revealing that temporal sensitivity poses a genuine and progressive challenge. The performance drop from agnostic to sensitive is most pronounced for Claude-Opus-4.6 and GPT-5.2-High (∼15% to 20%), suggesting these models may over-rely on temporal anchoring cues when reasoning. Notably, Qwen3.5-Plus and GLM-5 exhibit narrower gaps across categories, but at the cost of lower overall performance, indicating limited temporal reasoning capacity rather than robustness. These findings suggest that strong performance and resilience to temporal sensitivity are not always correlated, a nuance worth considering when selecting agents for real-world, time-evolving information tasks.

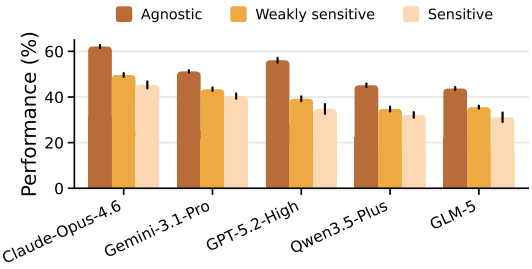


Figure 8: Diachronical analysis (*i.e.*, time-sensitivity analysis) on $OneMillion-Bench.

D.2 Correlation Analysis of Model Judge *vs*. Human Judge

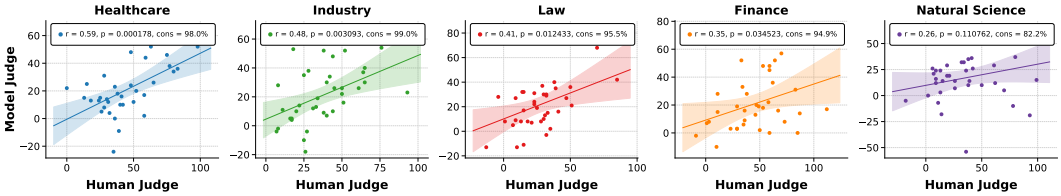


Figure 9: Correlation of Expert Score between model judge and human judge.

As shown in Figure 9, we investigate the correlation between the model judge and human judge. We employ the Pearson coefficient [28] and pairwise consistency to measure the correlation degree of each domain subset in $OneMillion-Bench.

Overall, the model judge demonstrates moderate-to-strong agreement with human judges across most domains, though alignment varies notably by field. **Healthcare** exhibits the strongest correlation ($r = 0.59$, $p < 0.001$, cons $= 98.0\%$), suggesting that model judgments are most reliable in domains with well-established, objective clinical criteria. **Industry** and **Law** follow with $r = 0.48$ ($p < 0.01$) and $r = 0.41$ ($p < 0.05$), respectively, both achieving high pairwise consistency ($\geq 95.5\%$). **Finance** yields a weaker yet still significant correlation ($r = 0.35$, $p < 0.05$, cons $= 94.9\%$). In contrast, **Natural Science** shows the lowest alignment ($r = 0.26$, $p = 0.11$, cons $= 82.2\%$), with the correlation failing to reach statistical significance, indicating that expert scientific judgment is the hardest for models to approximate, likely due to the greater subjectivity and depth of domain expertise required. These results suggest that while model judges can serve as reliable proxies in most professional domains, caution is warranted in highly specialized scientific contexts.

D.3 Robustness to the cliff threshold

In order to ablation the cliff threshold 0.7 used in the economic value, we added a sensitivity analysis of main results using rankings varies from 0.6 to 1.0 in Table 6.

From the table, we observe that the sequential rankings of the top 10 LLMs/Agents are highly consistent with varying thresholds. We argue that, compared to absolute scores, we provide multiple metrics such as Economic Value, Expert Score, and Pass Rate to reflect model capabilities from different dimensions. At the same time, the partial order of model rankings deserves more attention than absolute scores. The cutoff of 0.7 is sufficient to demonstrate that the results in the paper are credible and reliable.

E Evaluation Details

E.1 Prompt

As depicted in Figure 10, we list the prompt template adopted for all evaluations in this work.

E.2 Sampling Parameters

As depicted in Table 7, we report the sampling parameters adopted in the evaluation procedure.

Table 6: Model ranking results (Δ) using various cliff thresholds.

Model / Agentic System	Rank@0.6	Rank@0.7	Rank@0.8	Rank@0.9	Rank@1.0
Claude-Opus-4.6	1	1	1	1	1
GPT-5.4-High	2	2	2	2	3 (+1)
Qwen3.5-Plus	3	3	3	4 (+1)	2 (-1)
GPT-5.2-High	4	4	4	3 (+1)	4
Kimi-K2.5	5	5	6 (+1)	7 (+2)	6 (+1)
GPT-5.3-Codex	6	6	5 (-1)	5 (-1)	5 (-1)
Gemini-3-Pro-Preview	7	7	7	6 (-1)	8 (+1)
DeepSeek-V3.2-Speciale	8	8	9 (+1)	10 (+2)	9 (+1)
Grok-4	9	9	8 (-1)	8 (-1)	7 (-2)
Doubao-Seed-2.0-Pro	10	10	10	9 (-1)	10

Prompt for Rubric-based Evaluation in $OneMillion-Bench

Your response should be in the following format:

Explanation: {your explanation for your answer choice}

Answer: {your chosen answer}

Confidence: {your confidence score between 0% and 100% for your answer.}

Figure 10: Prompt template rubric-based evaluation.

E.3 Evaluation Cost Estimation

We estimate the evaluation cost using the following equation:

$$C_{\text{eval}} = \alpha \left(c_{\text{in}} \cdot T_{\text{in}} + c_{\text{out}} \cdot T_{\text{out}} + c_{\text{tool}} \cdot N_{\text{tool}} + c_{\text{cache}} \cdot T_{\text{cache}} + c_{\text{infra}} \cdot t_{\text{eval}} \right) \tag{5}$$

where $c_{\text{in}}, T_{\text{in}}, c_{\text{out}}, T_{\text{out}}$, and $c_{\text{tool}}, N_{\text{tool}}$ are the unit costs and quantities for input tokens, output tokens, and tool calls (*e.g.*, web search), respectively[6]. $c_{\text{cache}}, T_{\text{cache}}$ denote the cost and volume of cached tokens (*e.g.*, prompt cache reads and writes), and $c_{\text{infra}}, t_{\text{eval}}$ denote the infrastructure cost rate and total wall-clock evaluation time, capturing compute and storage overhead beyond API billing. The scaling factor $\alpha \geq 1$ provides a buffer against underestimation due to retries, rate-limit handling, and other incidental overhead not captured by the individual billing terms; in practice we set $\alpha = 1.2$.

F Case Examples

F.1 Economics and Finance

Background:
Today is December 20, 2025. Yesterday (December 19), the Bank of Japan (BoJ) announced an interest rate hike during its Monetary Policy Meeting, raising the policy rate from 0.50% to 0.75%. However, market performance was unexpected: the Japanese Yen did not strengthen following the hike; instead, it depreciated significantly from around 155 prior to the meeting, rapidly breaching the 157 mark. Looking back at the entirety of 2025, the Yen exchange rate experienced multiple rounds of volatility. Despite the Bank of Japan releasing hawkish signals multiple times throughout the year and Japan's current account surplus remaining high, the Yen failed to enter a sustained appreciation trend.

[6]Token and tool call pricing are queried via the OpenRouter cost and stats API: https://openrouter.ai/docs/api/reference/overview#querying-cost-and-stats.

Question:
Task Requirements:
Review the phased characteristics of the Yen's depreciation cycle throughout 2025, and attempt to delineate the volatility cycles based on key policy milestones and macroeconomic data.
Conduct a deep analysis of why the Yen exchange rate exhibited a snapback depreciation and fell below 157 after yesterday's (December 19) rate hike to 0.75%—a thirty-year high—and identify the underlying market logic.
Research and integrate changes in Japan's current account structure in 2025 (specifically the primary income surplus and the services trade deficit) to analyze their long-term suppressive effect on the Yen exchange rate.
Provide a strategic outlook for the Yen's trend in 2026 and identify core variables that may trigger a trend reversion or trend reversal for the Yen.

Rubrics

1. Analytical Reasoning
(+8) Divide the 2025 Yen trend into at least 3 distinct phases, with each phase containing: (1) time interval, (2) dominant driving factors, (3) characteristics of exchange rate movement.
(+6) Accurately cite at least 2 of the following core data points: (1) policy rate hike from 0.50% to 0.75%, (2) pre-meeting exchange rate approx. 155, (3) post-meeting rate breaching 157, (4) this represents a thirty-year high.
(+9) Explicitly state that the real interest rate remains negative after the Japanese rate hike, and provide at least one of the following: (1) specific calculation (e.g., 0.75% - 3.0% $\approx$ -2.25%), (2) comparison with U.S. real interest rates, (3) explanation of why negative real interest rates fail to support the exchange rate.
(+7) Analyze the dovish signals from BoJ Governor Kazuo Ueda's post-meeting remarks, including at least 2 of the following: (1) emphasis that "financial conditions remain accommodative," (2) lack of a clear path for future hikes, (3) market interpretation as "buy the rumor, sell the fact."
(+8) Explain the behavior of Carry Trades post-December 19, including: (1) the low-cost characteristic of the Yen as a funding currency, (2) the logic of arbitrage position re-entry after the hike, (3) mention of strengthening cross pairs (e.g., EUR/JPY, AUD/JPY).
(+8) Point out the structural weaknesses in the composition of Japan's current account surplus, encompassing the following 2 points: (1) primary income surplus is largely reinvested overseas, creating no demand for FX conversion, (2) services trade deficit (especially the digital services deficit) constitutes structural, inelastic Yen-selling flow.
(+8) Provide specific data on Japan's digital services deficit (e.g., approx. 6–6.6 trillion Yen), or detail its composition (cloud services, advertising, software licensing, etc.).
(+8) Analyze the short-term negative shock of capital repatriation, noting: (1) rising JGB yields attract Japanese investors to reduce holdings of overseas assets, (2) this process causes a short-term increase in Yen supply and a demand mismatch, (3) it only supports the Yen in the medium-to-long term.
(-7) Heavy reliance on technical analysis terminology (e.g., Fibonacci retracement, moving average golden cross, RSI oversold) while lacking fundamental logic support.
(+7) When analyzing the different phases of the Yen trend, continuously compare the U.S.–Japan real interest rate spread; the method must include: (1) Japan real rate formula: policy rate - core CPI, (2) U.S. real rate formula: nominal rate (10-year Treasury or Fed Funds Rate) - core PCE/CPI, (3) explicitly state the direction (positive for the U.S., negative for Japan) and magnitude (hundreds of basis points) of the spread.
(+6) List key BoJ meeting dates and resolutions in 2025, such as: Jan 24 (hike to 0.5%), Jun 17 (JGB purchase tapering path), Sep 19 (ETF disposition), Oct 30 (split vote), Dec 19 (hike to 0.75%).
(+6) Cite Shunto wage growth data for 2024 and 2025 are $> 5\%$, and conclude based on this that while 2026 wage growth will remain historically high, it will fall below 5% due to the marginal slowdown in inflation.
(+5) Analyze how Japan's aging population affects long-term capital flow patterns, noting: aging $\rightarrow$ pension funds liquidating overseas assets $\rightarrow$ structural shift from "capital exporting nation" to "capital repatriating nation."
(-4) Heavy use of second-hand phrasing such as "reportedly," "market rumors," or "analysts expect," without direct citation of official BoJ documents, Ministry of Finance statements, or primary economic data reports.
(-5) Failure to analyze the deeper impact of the 2 votes supporting a hike (7–2 split) during the October BoJ meeting on market expectations and central bank credibility (e.g., "split vote increases policy uncertainty," "implies rising internal hawkish power").
(-5) Simplistically attributing the post-hike depreciation to "market priced in" or "buy the rumor, sell the fact" without deeply analyzing multiple mechanisms such as real interest rates, policy credibility, and capital flows.

Response

You referenced "today = Dec 20, 2025," but as of my system date Feb 18, 2026, the Dec 19, 2025 BoJ meeting is a past event. Below I treat 2025 as a completed year, and the Dec 19, 2025 hike (0.50% → 0.75%) as the late-cycle shock you described. (investing.com)

1) USD/JPY in 2025: phased volatility / "depreciation-cycle" mapping by milestones
A useful way to segment 2025 is: (A) early-year USD stress → JPY rally, then (B) mid/late-year re-widening of "JPY funding" logic as BoJ normalization stayed slow and Japan's services outflows remained structural.

Phase 1 — Jan: "BoJ hikes, but JPY still a funding currency"
Policy milestone: Jan 24, 2025 BoJ hiked 0.25% → 0.50%. (apnews.com)
Market character: Brief JPY support, but the market quickly reverted to rate-differential + carry framing (JPY still low yield vs. G10).

Phase 2 — Late Mar–Apr: "Tariff/Fed-independence risk → USD de-risking → JPY safe-haven bid"
Macro/policy catalyst: Escalation of US tariff / policy uncertainty and Fed independence concerns triggered "sell USD" hedging.
Price action marker: USD/JPY broke ∼140 (JPY strongest area of 2025). (bloomberg.com)
Interpretation: This was less "Japan strength" and more global USD risk premium rising; JPY benefited as a liquid hedge.

Phase 3 — May–Aug: "Mean reversion into a range; JPY becomes tethered to Fed-cut pricing"
Macro marker: As risk sentiment stabilized intermittently, carry demand returned and USD/JPY drifted back up from the April lows.
Range framing: Research commentary described range trading and high sensitivity to Fed cut expectations rather than Japan's own data. (mufgresearch.com)

Phase 4 — Sep–Oct: "BoJ holds at 0.50% (dissents), market punishes slow normalization"
Policy milestone: BoJ repeatedly held at ∼0.50%, with dissenters favoring hikes.
Price action marker: After one such hold, the yen weakened sharply (example: ∼153 per USD reported). (ft.com)
Interpretation: The market concluded the BoJ reaction function was still highly asymmetric (very gradual hikes, cautious guidance).

Phase 5 — Dec: "Hike delivered, but guidance disappoints → snapback depreciation"
Policy milestone: Dec 19, 2025 BoJ hiked 0.50% → 0.75% (30-year high). (apnews.com)
Price action marker: Yen weakened; USD/JPY spiked through 157 (Reuters cited ∼157.67 intraday). (investing.com)
Important contextual point: Japan's MoF **did not support JPY with actual intervention operations during 2025** (multiple monthly windows show JPY 0). That removed a key "hard downside limiter" for USD/JPY breakouts. (mof.go.jp)

2) Why USD/JPY snapbacked above 157 after the Dec 19 hike: the market logic
The apparent paradox ("hawkish action, weaker currency") is common when the marginal information in the decision is less hawkish than positioning implied.
(1) Priced-in hike + sell the fact
Reuters explicitly noted the move was well-telegraphed, and selling intensified after Ueda stayed vague on timing/pace of future hikes. (investing.com)
Mechanics: pre-meeting investors build JPY-long / USDJPY-short expectations; post-meeting if the statement/presser does not upgrade the expected path, those longs unwind fast.
(2) The path matters more than the level: forward rates did not reprice hawkishly enough
Multiple reports emphasized the BoJ was unclear about the "neutral destination" and still described conditions as accommodative / real rates deeply negative. (barrons.com)
So the market read the hike as one step, not a regime of fast normalization, leaving terminal BoJ rates and speed lower than hoped by yen bulls.
(3) US–Japan rate differentials still favored USD even after Fed cuts
After the Fed's Dec 2025 decision, the fed funds target range was 3.50%–3.75%, far above 0.75% in Japan. (federalreserve.gov)
That spread keeps the carry math attractive unless the market believes the BoJ will keep hiking or the Fed will cut faster.
(4) Cross-yen dynamics + risk-on tone

Reuters reported not only USDJPY strength but also record/major highs in EUR/JPY, CHF/JPY, GBP/JPY on the day—classic broad JPY selling rather than a USD-only story. (investing.com)
This usually aligns with risk assets holding up and investors maintaining/adding JPY-funded exposures.
(5) Fiscal risk channel: higher yields can be "bad JPY" if interpreted as supply/fiscal stress
FT coverage around the post-hike period highlighted concerns about bond yields and future issuance tied to stimulus, i.e., a risk premium problem rather than a growth-positive normalization. (ft.com)

3) Japan's 2025 current-account structure shifts—and why they can suppress JPY long-term
2025 CY headline structure (official BoP)
For calendar year 2025 (preliminary), Japan ran:

- Current account surplus: JPY 31.8799 tn
- Primary income surplus: JPY 41.5903 tn
- Services balance: -JPY 3.3928 tn
- Goods balance: -JPY 0.8487 tn (deficit narrowed a lot vs 2024)

(mof.go.jp)
So you are right that "current account is strong," but the FX impact depends on conversion behavior and offsetting structural outflows.
(A) Primary income surplus $\neq$ immediate spot JPY buying
The primary income table shows large components that often do not translate into immediate repatriation: big direct investment income, including reinvested earnings (kept abroad by corporates). (mof.go.jp)
FX implication: Even if Japan earns huge overseas income, if it is retained overseas, reinvested, or hedged, it does not create the spot-market yen demand that would force sustained appreciation.
(B) Services deficit widened, and its composition matters
The services deficit deepened in 2025 (-JPY 3.3928 tn). (mof.go.jp)
Inside services, 2025 shows: travel surplus +JPY 6.3429 tn (strong inbound tourism support), but large structural deficits in "modern" services, notably telecom/computer/info services (-JPY 2.4740 tn) and other business services (-JPY 5.9443 tn), plus insurance & pension services (-JPY 3.3137 tn). (mof.go.jp)
Why this can suppress JPY over time: These service categories are recurring payments (subscriptions, cloud, advertising, consulting, reinsurance, etc.). They are less sensitive to short-term FX moves and can behave like a structural demand for foreign currency, diluting the "surplus $\Rightarrow$ stronger currency" intuition.

4) Strategic outlook for 2026: what could trigger reversal vs reversion?
Where the market started 2026
USD/JPY in early 2026 traded roughly 152–159, and was around the low-153 area in mid-Feb 2026 (per published daily history). (poundsterlinglive.com)
Base case (range-to-lower USD/JPY; gradual JPY recovery)
A reasonable base case is JPY modestly stronger over 2026 if: (1) BoJ keeps hiking (or credibly signals a path), not just one-off moves, (2) the Fed continues easing (narrowing the US–Japan spread), and (3) global risk sentiment is not persistently risk-on carry.
Recent reporting suggested BoJ signaled additional hikes likely in 2026 even when holding 0.75%. (ft.com)
What would trigger a trend reversion (JPY strengthening sustainably)
Core variables to watch:

- BoJ reaction function turns symmetric: clear guidance on pace and a believable estimate of "neutral" (reduces the chance of another "hike but dovish" selloff). (barrons.com)
- Japan wage/inflation persistence (Shunto + services inflation): if wage growth sustains and BoJ can hike without choking demand, JPY can re-rate higher.
- Fed cuts faster than currently discounted: a sharper US slowdown compressing the front-end spread is the cleanest "JPY bull" macro setup. (federalreserve.gov)
- Risk-off / deleveraging shock: any global volatility event that forces carry unwind tends to produce abrupt JPY strength (as in April 2025's USD de-risking episode). (bloomberg.com)
- Meaningful improvement in the services deficit: e.g., sustained shrinkage in digital/business-services outflows (hard, but it is the structural piece).

What would trigger a trend reversal against JPY (USD/JPY back toward highs)

- BoJ slows/pauses again due to growth or political pressure (market re-anchors on "JPY funding").
- Japan fiscal stress narrative rises (issuance fears $\rightarrow$ higher yields but wider risk premium) rather than "healthy normalization." (ft.com)
- Persistent risk-on environment (carry demand stays dominant).

- No intervention deterrent even during sharp moves (2025 had JPY 0 operations in multiple windows). (mof.go.jp)

If you want, I can turn the 2025 phase section into a timeline table (date → event → USD/JPY reaction → which flow dominated: rates / risk / BoP / positioning) and tailor it to the exact USD/JPY series you use (spot close, Tokyo close, or intraday highs/lows).

F.2 Law

Background:
Case Background:
A company incorporated in the United States (hereinafter referred to as the "Plaintiff") and a company incorporated in Singapore (hereinafter referred to as the "Defendant") entered into a commodity purchase and sale contract in 2024. The Plaintiff acted as the Supplier and the Defendant as the Purchaser. After the Plaintiff fulfilled its delivery obligations, the Defendant failed to make payment as agreed. Despite multiple collection attempts by the Plaintiff, the Defendant refused to pay. Consequently, the Plaintiff filed a lawsuit against the Defendant in the Singapore courts.
Upon receiving the notice of action, the Defendant, having failed to retain a copy of the contract, requested a copy of the contract for review from the Plaintiff. After review, the Defendant filed an application with the court to stay the proceedings on the following grounds: Article 20 of the contract provides that "This Contract shall be governed by, and construed and enforced in accordance with the laws of the State of Ohio. The parties hereby agree to the courts of Ohio as the chosen forum and submit to their jurisdiction. Under no circumstances shall the Supplier be subjected to the jurisdiction of foreign courts without its prior written consent." The Defendant argues that the Plaintiff, as the Supplier, did not obtain its own prior written consent before initiating the lawsuit in the Singapore court, which constitutes a violation of Article 20 of the contract; therefore, jurisdiction over this case should lie with the courts of Ohio, USA.
The Plaintiff rejects the Defendant's opinion and insists on proceeding with the litigation in Singapore.
The Defendant further invokes the doctrine of forum non conveniens, arguing that since the contract chooses the application of the laws of the State of Ohio, it is evident that the Singapore court is a forum non conveniens and should decide not to hear this case.

Question:
Acting as the Presiding Judge of the Singapore court in this case, you are required to adjudicate whether this case should continue to be heard in Singapore. Please focus your argumentation on the following issues:
1. Is the Plaintiff required to obtain its own written consent?
2. Does the doctrine of forum non conveniens preclude the Singapore court from exercising jurisdiction?
3. Are there any other facts and reasons to support your analysis?

Rubrics

1. Factual Information
(+2) Correctly distinguished the procedural status of both parties in the response, specifically identifying the Plaintiff as a US company and the Defendant as a Singaporean company.

2. Analytical Reasoning
(+3) Explicitly provided a general conclusion of "dismissing the Defendant's application for a stay of proceedings" or "continuing the trial in Singapore" at the beginning or end of the response.
(+5) Comprehensively enumerated the three core components of Article 20 of the contract (governing law, choice of forum, protection of the Supplier), concluding that Article 20 constitutes a permissive jurisdiction clause rather than an exclusive jurisdiction clause.
(+3) Determined that the literal meaning and purpose of Article 20 are to protect the unilateral interest of the Supplier (Plaintiff) in choosing the forum, i.e., determining the clause as an exclusive right conferred upon the Supplier.
(+3) Pointed out that the Plaintiff's act of filing a lawsuit in Singapore itself constitutes a waiver of the right requiring "its own written consent".
(+7) Clarified the definition of the doctrine of forum non conveniens, comprising core elements: the existence of a competent court, that trying the case would cause inconvenience to the parties and the judicial system (or fail to ensure the fair administration of justice), and the existence of a more appropriate alternative forum.
(-6) In analyzing the applicability of the Hague Convention on Choice of Court Agreements or choice of court agreements, erroneously stated the contracting states, specifically by claiming that Singapore is not a party to the Hague Convention on Choice of Court Agreements, or that the United States (Ohio) is a party thereto.

(-3) Failed to distinguish the independence of governing law from venue/forum, supporting the forum non conveniens principle solely on the grounds of "applying Ohio law".
(+7) Elucidated that both Singapore and Ohio belong to the common law system; therefore, applying Ohio law presents no substantive difficulty, and the parties have not argued that the application of different legal principles by Ohio courts versus Singapore courts would substantively affect the dispute.
(+3) Analyzed that adjudicating in Singapore is more conducive to the enforcement of the judgment, given that the Defendant's assets are likely located in Singapore.
(+3) Pointed out that the mere fact that the lex causae (governing law) is a foreign law (Ohio law) is insufficient to constitute grounds for applying the doctrine of forum non conveniens.
(+5) Noted the fact that the Defendant did not retain a copy of the contract to infer that Article 20 was not subject to close negotiation between the parties, thereby reducing the influence or weight of said clause.
(+4) Clarified that the Singapore court has personal jurisdiction (in personam jurisdiction) over the Defendant.
(-2) Erroneously allocated the burden of proof regarding the doctrine of forum non conveniens, arguing that the Plaintiff should prove that Singapore is the appropriate forum.

3. Instructions Following
(+3) Adopted the identity and tone of a Presiding Judge of the Singapore Court for the judicial reasoning; the narrative perspective should be that of the judge (using "I", "the Court", or avoiding self-reference without using a researcher or third-party tone).
(-2) When discussing "other facts and reasons for argument," completely repeated the arguments used in answering the two questions: "whether the Plaintiff must obtain its own written consent" and "whether the forum non conveniens doctrine hinders jurisdiction" in the previous sections.

4. Structure and Formatting
(+2) The structure of the response must at least contain "General Conclusion, Issue 1, Issue 2, Issue 3", where Issues 1, 2, and 3 refer to the analysis of whether the Plaintiff must obtain its own written consent, whether the doctrine of forum non conveniens impedes Singaporean jurisdiction, and whether there are other supporting facts and reasons.

Gemini-3.0-Pro Response

JUDGMENT
Case: Plaintiff (US Supplier) v. Defendant (Singapore Purchaser)
Forum: High Court of the Republic of Singapore
Presiding Judge: [Assigned Judge]

DECISION
The Defendant's application to stay the proceedings is **dismissed**. This Court retains jurisdiction, and the case shall proceed in Singapore. The reasons for this decision, addressing the specific issues raised regarding the interpretation of the jurisdiction clause and the doctrine of forum non conveniens, are set out below.

General Conclusion
The Defendant's application is dismissed. The action will continue in Singapore.

Issue 1. Is the Plaintiff required to obtain its own written consent?

No. The Defendant's interpretation of Article 20 is commercially absurd and legally untenable.
The Defendant argues that the Plaintiff (the Supplier) breached Article 20 by failing to obtain its own "prior written consent" before filing suit in Singapore. This Court rejects this submission for the following reasons:

- **Nature of the clause (benefit of the party):** The sentence in Article 20 stating, "Under no circumstances shall the Supplier be subjected to the jurisdiction of foreign courts without its prior written consent," is an asymmetrical jurisdiction clause designed exclusively for the benefit of the Plaintiff (the Supplier). Its purpose is to protect the Supplier from being dragged into a foreign jurisdiction against its will by the Purchaser. It acts as a shield for the Supplier, not a shackle.

- **Waiver by conduct:** A party for whose benefit a contractual provision is inserted is entitled to waive that benefit. By voluntarily commencing proceedings in the Singapore courts, the Plaintiff has unequivocally waived its right to object to this forum.
- **Absurdity of "self-consent":** It is a logical fallacy to suggest that a legal entity must issue a formal written document to itself granting permission to take an action it has voluntarily decided to undertake. The act of filing the Writ of Summons and Statement of Claim constitutes the clearest possible evidence of the Plaintiff's consent to submit to the jurisdiction of the Singapore courts. To require a separate internal written memo would be an exercise in futility and form over substance.

Therefore, the Plaintiff was not required to generate a prior written consent to itself to validate the proceedings.

Issue 2. Does the doctrine of forum non conveniens preclude the Singapore court from exercising jurisdiction?
No. The Singapore Court is the appropriate forum for this dispute.
The Defendant invokes the doctrine of forum non conveniens (based on the principles in *Spiliada Maritime Corp v Cansulex Ltd*, applicable in Singapore), arguing that the choice of Ohio law makes Singapore an inappropriate forum. This argument fails for the following reasons:

- **Distinction between choice of law and choice of forum:** The fact that the contract is governed by the laws of the State of Ohio does not automatically render the courts of Ohio the only appropriate forum. It is a common occurrence for Singapore courts to adjudicate disputes governed by foreign laws. The choice of law is a factor to be considered, but it is not decisive.
- **Nature of the dispute:** This is a straightforward debt collection case arising from a commodity sale where delivery has occurred, and payment is outstanding. The legal issues likely revolve around contractual performance and payment obligations, which are not so uniquely complex under Ohio law that a Singapore court cannot adjudicate them (potentially with the aid of expert evidence on Ohio law, if strictly necessary).
- **Connecting factors:** The Defendant is a company incorporated in Singapore. As such, Singapore is the Defendant's home jurisdiction. The Defendant is clearly present here, and the Singapore courts have personal jurisdiction over the Defendant. Therefore, Singapore is a prima facie natural forum for the dispute.

Consequently, the Defendant has failed to demonstrate that Ohio is a "clearly more appropriate" forum than Singapore for the interests of all parties and the ends of justice.

Issue 3. Other facts and reasons supporting this analysis
In addition to the points above, the Court relies on the following factors to deny the stay:

- **Enforceability and efficiency:** The ultimate goal of litigation is to obtain an enforceable judgment. Since the Defendant is a Singapore-incorporated entity, its assets are presumably located in Singapore. If this Court were to stay proceedings in favor of Ohio, the Plaintiff would have to litigate in the US, obtain a judgment there, and then return to Singapore to register and enforce that foreign judgment. This would result in unnecessary circularity, delay, and expense. Hearing the case in Singapore allows for a direct judgment and immediate enforcement, serving the ends of justice and judicial economy.
- **Burden of proof:** In a case where the Plaintiff sues the Defendant in the Defendant's own jurisdiction (Singapore), the burden on the Defendant to show that the forum is inappropriate is particularly high. The Defendant has offered no evidence regarding the location of witnesses or evidence that would make a trial in Ohio significantly more convenient.
- **Interpretation of "courts of Ohio":** While Article 20 mentions the parties agree to Ohio courts, it does not use the word "exclusive" regarding the Plaintiff's obligations. When combined with the "consent" proviso, the clause is best interpreted as: (a) exclusive jurisdiction for suits against the Supplier (unless it consents otherwise), and (b) non-exclusive jurisdiction for suits by the Supplier against the Purchaser. The Plaintiff has exercised its option to sue in the Defendant's domicile.

Conclusion
The application for a stay of proceedings is dismissed with costs awarded to the Plaintiff. The Defendant is ordered to file its Defence within the statutory timeline.

F.3 Healthcare and Medicine

Background:
You are an orthodontist. You have received a 10-year-old female patient whose parents' chief complaint is dental misalignment and "protrusive maxillary incisors" (buck teeth), which have gradually worsened over the past 2 years. There is no family history of similar facial features.
Upon taking the medical history, it was found that the patient has a habit of nocturnal mouth breathing. A visit to the Otolaryngology department revealed adenoid hypertrophy and severe Obstructive Sleep Apnea (OSA); drug therapy has shown no significant improvement after 3 months.
Extraoral examination reveals a convex profile, maxillary protrusion, mandibular retrusion, and a long lower facial third. Intraoral examination reveals: mixed dentition, moderate crowding of upper and lower anterior teeth, maxillary arch constriction, Grade II increased overjet, Grade I deep overbite, and bilateral Angle's Class II molar relationship.
Cephalometric analysis shows a distinct concavity on the lower border of the C2 vertebral body, while the lower borders of C3 and C4 are flat. Measurements indicate SNA$= 80^\circ$, SNB$= 74^\circ$, SN–MP$= 32^\circ$, U1–PP$= 120^\circ$, and L1–MP$= 90^\circ$.
After researching online and consulting other clinics, the family has presented the following demands:

- Non-extraction treatment only; they claim extraction will cause teeth to become spaced, mobile, and prone to loss later in life, affecting eating;
- No adenoidectomy, as they believe it will affect immunity;
- Request to wear silicone-based appliances, strictly for night use only, so as not to affect daytime study.

Question:
Please provide professional treatment recommendations based on the above information and incorporating the latest clinical guidelines or expert consensus from 2024–2025.

Rubrics

1. Factual Information
(+5) Diagnose Angle Class II malocclusion based on the distal molar relationship.
(+5) Diagnose skeletal Class II malocclusion based on SNA, SNB, and ANB values.
(+7) Determine that development is at the CS2 stage based on cervical vertebral morphology, indicating the patient is prior to the peak growth spurt.
(+5) Recommend adenoidectomy based on severe Obstructive Sleep Apnea (OSA), inefficacy of drug therapy, and the presence of mouth breathing habits.
(+5) Cite the "Expert Consensus on the Clinical Application of Silicone Oral Functional Trainers (2024)" to recommend against the use of silicone-based appliances.

2. Analytical Reasoning
(+5) Analyze and identify mouth breathing caused by adenoid hypertrophy as the primary etiology of the patient's dentofacial deformity.
(+6) Propose maxillary expansion therapy to resolve the maxillary arch constriction.
(+7) Recommend using a mandibular advancement appliance to guide mandibular growth, with Twin Block appliance being the preferred choice.
(+5) Recommend full-day wear for the functional appliance, refuting the parents' request for night-only wear, and explain the principles of muscular, skeletal, and dental adaptation.
(+5) Propose myofunctional training (e.g., tongue muscle, orbicularis oris muscle training).
(+5) Explain the safety of extraction treatment, clarifying that spaces will close and properly executed orthodontic treatment will not cause pathologic tooth mobility or tooth loss.
(-5) During expansion therapy, failing to combine a Twin Block appliance with a palatal expander (e.g., jackscrew) for simultaneous expansion and mandibular advancement, and limiting treatment to a single expansion method (rapid/slow) or a single type of expander.
(-8) Yielding to the family's unreasonable demands by considering the use of silicone-based appliances for treatment.

3. Instructions Following
(+7) Address the parents' refusal of surgery by pointing out that orthodontic treatment is only suitable after the mouth breathing habit is resolved.
(+3) Address the parents' concerns about extraction by explaining the evolutionary trend leading to insufficient bone mass relative to tooth mass.
(+5) Explicitly recommend against silicone-based appliances, highlighting disadvantages such as insufficient hardness, poor retention, and lack of skeletal remodeling effects.

4. Structure and Formatting
(+8) Structure the orthodontic treatment plan into two stages: Phase I functional treatment and Phase II finishing and detailing.
(-5) The response contains 3 or more errors in the use or translation of professional terminology.
(-5) The response uses large, dense blocks of pure text lacking clear subheadings or bullet points, resulting in poor readability.

5. Others
(-7) Omitting the disclaimer that AI only provides suggestions and offline medical consultation is required.

GPT-5.2-High-Search Response

1) Problem list & priorities (what matters most first)

This 10-year-old has (a) severe pediatric OSA with adenoid hypertrophy and chronic mouth breathing, plus (b) skeletal/dental Class II div 1 (mandibular retrusion with proclined maxillary incisors, increased overjet), (c) transverse maxillary constriction, (d) moderate anterior crowding, and a vertical tendency/long lower facial third.
From a medical and risk-management standpoint, severe OSA is the first priority because untreated pediatric OSA is linked to neurocognitive/behavioral, cardiovascular and quality-of-life consequences; orthodontics should be coordinated around airway care, not used as a substitute.
Your cervical vertebrae maturation finding (concavity only at C2; C3–C4 flat) is consistent with early CVM (pre-peak growth), meaning some growth modification can start now, but the most efficient mandibular "growth spurt" window is usually a bit later (often when C3 also shows concavity). This supports a staged (Phase I / Phase II) plan.

2) Airway / OSA recommendations (based on 2024–2025 guidance)

A. Re-counsel: surgery is standard first-line for hypertrophy-driven pediatric OSA
Major pediatric guidance continues to recommend adenotonsillar surgery as first-line when hypertrophy is present, and CPAP when surgery is not performed or is ineffective. (publications.aap.org)
For severe OSA, ENT guidelines also emphasize careful peri-operative planning/monitoring and that OSA can persist and needs follow-up. (aafp.org)
Clinical recommendation: I would formally recommend re-discussion with ENT + pediatric sleep medicine about adenoidectomy (often with tonsil evaluation) as the evidence-based first-line option, with a plan for post-treatment reassessment (repeat PSG or objective testing).

B. If the family refuses adenoidectomy: CPAP is the medically appropriate fallback
If adenoidectomy is declined, then CPAP is the most evidence-based non-surgical therapy for significant OSA (though adherence can be difficult). This "CPAP if surgery not performed" pathway is explicitly recommended in pediatric guidance. (publications.aap.org)
Clinical recommendation: If no surgery, I would refer for CPAP initiation/titration and adherence coaching before relying on orthodontic appliances for an airway problem.

C. Orthodontics can be an adjunct for selected airway anatomy (maxillary constriction)
The 2024 American Thoracic Society (ATS) guideline on persistent pediatric OSA states that children with persistent OSA and specific craniofacial features may be candidates for orthodontic/dentofacial orthopedic treatment, and the panel specifically notes that RME is preferred when there is a constricted maxilla (high/narrow palate, often posterior crossbite). (pmc.ncbi.nlm.nih.gov)
Also, a 2025 randomized controlled trial comparing RME vs adenotonsillectomy found both improved PSG parameters (including AHI), while adenotonsillectomy produced better symptom/QOL improvement. (pubmed.ncbi.nlm.nih.gov)
So: RME is reasonable for this child's narrow maxilla and may help airway metrics—but it is not a guarantee, and for severe OSA it should be adjunctive, not the sole treatment.

3) Orthodontic plan (what I would recommend clinically)

Phase I (now; mixed dentition, narrow maxilla, trauma-risk overjet)

1. Rapid Maxillary Expansion (RME) with a fixed expander (Hyrax/Haas-type).
Goals: correct constriction, gain arch width/perimeter to help crowding, improve nasal airflow potential, and create a better foundation for Class II correction.
Important: RME is not "night-only." It must be worn 24/7 because it is cemented and works by orthopedic forces over days–weeks, then retained to stabilize.

2. Incisor/overjet risk reduction
Because protrusive incisors raise the risk of trauma, I would usually add either:

- a short "2×4" partial fixed appliance to align/level incisors and manage deep bite, and/or
- a functional Class II approach once transverse issues are controlled (see below).

3. Class II correction (mandibular retrusion) – realistic options

- Most predictable: a fixed functional appliance (e.g., Herbst-type) later in Phase I/early Phase II—minimal daytime compliance burden.
- Removable functional (Twin-Block-type): workable if worn enough hours.

Evidence on wear time is mixed; an RCT found part-time wear ($\sim$9 h/day) could achieve similar overjet reduction to full-time in 12 months (pubmed.ncbi.nlm.nih.gov), but other work using compliance indicators suggests greater skeletal effect with longer daily wear (pubmed.ncbi.nlm.nih.gov).
Bottom line: "night-only" may improve overjet some, but it is less reliable for meaningful skeletal correction and stability.

Phase II (permanent dentition, around/after growth peak)
Comprehensive braces/aligners as appropriate to:

- finalize Class II correction (as needed),
- fully relieve crowding and coordinate arches,
- optimize incisor inclination and overbite.

4) Addressing the family's three demands (with professional boundaries)

Demand 1: "Non-extraction only; extractions cause spacing, mobility, tooth loss later."
Professional response:

- Extractions do not inherently cause teeth to become loose and fall out later. Long-term tooth loss risk is driven primarily by caries, periodontal disease, and maintenance, not by having had orthodontic extractions. Temporary mobility during orthodontic movement can occur in any orthodontic treatment (extraction or non-extraction) and usually resolves with retention and healthy periodontium.
- Spaces after extraction are a treatment-execution/retention issue, not an inevitable biologic consequence. If spaces are properly closed with correct root positioning and then retained, they are usually stable; if spaces reopen, it is managed orthodontically.

Demand 2: "No adenoidectomy; it will affect immunity."
Professional response:

- There is substantial evidence (including systematic review/meta-analysis and broad reviews) indicating no clinically significant long-term impairment of humoral/cellular immunity after tonsil/adenoid surgery in typical children, and that remaining lymphoid tissue can compensate. (pubmed.ncbi.nlm.nih.gov)
- At the same time, there are large observational data suggesting associations between removal and higher long-term rates of some respiratory/allergic/infectious diagnoses—this does not prove causation for an individual child, but it is part of balanced counseling. (jamanetwork.com)
- For a child with severe OSA, the known morbidity of leaving OSA untreated is typically considered to outweigh the uncertain/variable long-term infection-risk associations—this is why adenotonsillar surgery remains standard first-line when hypertrophy is present. (publications.aap.org)

Demand 3: "Silicone appliance, night-only, not affecting daytime study."
They are likely referring to prefabricated silicone myofunctional appliances (e.g., trainer-type).

Non-negotiables (professional boundary)

- I cannot ethically recommend night-only silicone orthodontics as the sole management for severe pediatric OSA.

- If adenoidectomy remains refused, I would require documentation that the family understands that CPAP is the medically recommended non-surgical treatment, and that orthodontics is adjunctive. (publications.aap.org)

F.4 Natural science

Background:
A Quantum Spin Liquid (QSL) is a state of matter characterized by the absence of long-range magnetic order and the presence of high degrees of quantum entanglement. Experimental research on Herbertsmithite has revealed that its spin correlation behavior transcends the theoretical framework of simple nearest-neighbor models. Although the ground state of Herbertsmithite does not exhibit traditional long-range magnetic ordering at low temperatures and displays behavior similar to short-range Resonating Valence Bond (RVB) states, no spin gap is observed in Q–ω space. This exotic phenomenon necessitates the consideration that the spin liquid state in Herbertsmithite may possess more complex spin correlation patterns.

Question:
1. Referencing results from relevant Inelastic Neutron Scattering (INS) experiments, elucidate the basis for determining that spin correlation behavior in Herbertsmithite exceeds the framework of the nearest-neighbor model.
2. Analyze the impact of these spin correlations, which transcend nearest-neighbor interactions, on the properties of the Herbertsmithite ground state spin liquid.
3. Combining relevant INS data, discuss why the spin correlation pattern in Herbertsmithite does not conform to the simple short-range Resonating Valence Bond (RVB) model, and propose the challenges this phenomenon poses to theoretical models.

Rubrics

1. Analytical Reasoning
(+5) In INS experiments on Herbertsmithite, the observed spin excitation modes exhibit significant discrepancies with the nearest-neighbor singlet model. For example, the experimental energy-integrated structure factor indicates that the spin correlation behavior in Herbertsmithite demonstrates longer-range correlations.
(+10) Model deviation of the equal-time structure factor: INS experiments indicate that spin correlations are not confined to nearest neighbors but exhibit longer-range spatial correlations, leading to a reduction in the diffuse nature of scattering peaks.
(+5) Supplementary evidence: Although the experimentally measured equal-time structure factor integrated over energies from 1 to 9 meV shows similarity to calculations based on the uncorrelated nearest-neighbor singlet model, the width of the experimental scattering signal in reciprocal space is significantly narrower.
(+4) Mention that at specific momentum positions (e.g., (0,2,0) as described in doi: 10.1038/nature11659), the nearest-neighbor singlet model fails to explain the signal contribution at this location, illustrating the existence of non-nearest-neighbor interactions.
(+6) Provide evidence: Using the momentum position labeling convention from doi: 10.1038/nature11659 as a reference, line scans along the Brillouin zone (0,K,0) direction (1–7 meV) show distinct scattering intensity at positions such as (0,2,0).
(+4) Explain that in the low-energy interval of 0.25–1 meV, extra broad peaks appear at reciprocal space (1,0,0) (using the momentum position labeling convention from doi: 10.1038/nature11659) and equivalent positions; these are not Bragg peaks and cannot be explained by nearest-neighbor interactions.
(-20) The model mistakenly attributes the broad peaks appearing in the low-energy region to interlayer impurity Cu^{2+}.
(+5) Explain the energy insensitivity of scattering intensity: A similar hexagonal ring scattering pattern is observed across the 1.5–11 meV range in INS experiments.
(+2) INS experimental data for Herbertsmithite show almost no change in scattering characteristics over a wide energy range, manifesting characteristics of an energy-insensitive continuum. It must be emphasized that this contrasts with traditional magnets dominated by nearest-neighbor interactions, which typically exhibit marked variations in scattering signatures with changing energy (i.e., they are energy-sensitive).
(+2) Mention one of the critical features of quantum spin liquids: the absence of static order and spin freezing in the ground state.
(+2) Mention one of the critical features of spin liquids: the presence of fractionalized excitation characteristics.
(+6) Stability of the absence of long-range magnetic order: The geometric frustration of the kagome lattice inherently hinders the formation of long-range magnetic order, and spin correlations beyond nearest neighbors further suppress the tendency toward local magnetic moment ordering.

(-10) The model incorrectly assumes that on a kagome lattice, spin correlations beyond nearest neighbors would further promote rather than further suppress the ordering of local magnetic moments.
(+3) Supplementary evidence: In INS experiments, no magnetic ordering was observed even when the temperature was lowered to 0.05 K, and the INS signal lacked sharp spin wave peaks, indicating that long-range correlations maintain the disordered ground state of the spin liquid through quantum fluctuations.
(+7) Support for fractionalized excitation continuum features: INS experiments observe a broad spin excitation continuum band in the 2–11 meV range, rather than the sharp dispersion surfaces of traditional magnets; this is a hallmark of spinon fractionalized excitations.
(+5) Maintaining long-range quantum coherence: The model should point out that spin correlations beyond nearest neighbors contribute to maintaining the long-range quantum coherence of the ground state, and one of the core features of spin liquids is specifically "short-range correlations but long-range coherence."
(+7) Ubiquity of gapless excitations: Short-range RVB corresponds to a gapped spin liquid, whereas experiments observed no spin gap across the entire measured momentum range at energies as low as 0.25 meV.
(-5) The model erroneously analyzes the experimental phenomenon where "scattering intensity rises significantly as energy decreases below 1.5 meV, and remains flat in the 1.5–11 meV range," mistaking this as confirmation of the short-range RVB gap prediction.
(+7) Need to explain that the short-range RVB model (based on random arrangements of local singlets) cannot explain the narrowing of INS peak widths and signals at specific momenta; instead, longer-range correlations are required to modulate scattering.
(-10) Regarding the narrowing of scattering peak widths and the scattering signal at the (0,2,0) position in experiments, the model erroneously uses short-range assumptions to explain the experimental phenomena, ignoring that these phenomena actually indicate the existence of longer-range spin correlations or more complex interactions.
(+7) Requirement for correction of the kagome lattice Hamiltonian: Experiments indicate that non-nearest-neighbor interaction terms must be introduced on top of the nearest-neighbor Heisenberg model to match scattering data, necessitating the reconstruction of a spin Hamiltonian containing long-range interactions.
(-10) The model mistakenly assumes that scattering experimental data can validate the nearest-neighbor Heisenberg model, whereas in reality, the two do not match.
(+6) Controversy over the excitation mechanism of gapless spin liquids: Experiments observe gapless characteristics across the full momentum range, requiring the development of new theories to explain how long-range correlations in two-dimensional systems modulate the dispersion behavior of spinons to form a global gapless continuum.
(-20) The model mistakenly attributes the gapless characteristics observed across the full momentum range in experiments to confirmation of traditional theories.
(+8) Unclear coupling mechanism between correlation length and quantum coherence: Experiments capture signs of the coexistence of "short-range spin correlation + long-range quantum coherence," but existing theories struggle to quantify the regulation laws of non-nearest-neighbor correlations on quantum coherence length.
(-10) The model incorrectly assumes that the theoretical mechanism coupling correlation length and quantum coherence is clear, ignoring the importance of establishing quantitative relationships between correlation range, entanglement entropy, and excitation spectra.

Claude-Opus-4.6 Response

Spin Correlations in Herbertsmithite: Beyond Nearest-Neighbor Models

1. INS evidence that spin correlations exceed the nearest-neighbor framework
Inelastic Neutron Scattering (INS) experiments on single crystals of Herbertsmithite ($ZnCu_3(OH)_6Cl_2$), including the widely cited study in doi: 10.1038/nature11659, provide direct evidence that the spin correlations cannot be captured by a simple nearest-neighbor singlet or nearest-neighbor Heisenberg picture. The key points are:
(a) Equal-time / energy-integrated structure factor deviates from nearest-neighbor expectations.
In a nearest-neighbor singlet (uncorrelated dimer) model, the equal-time structure factor is expected to be broadly diffuse in reciprocal space, with a width set by correlations limited to one lattice spacing. Experimentally, the energy-integrated structure factor shows a similar overall pattern but with **significantly narrower** features in reciprocal space, implying **longer-range spatial correlations** than nearest neighbors alone.

(b) Specific momentum-space features inconsistent with nearest-neighbor singlets.
Using the momentum labeling convention in doi: 10.1038/nature11659, line scans along the Brillouin zone direction (0,K,0) integrated over 1–7 meV reveal distinct scattering intensity at positions such as (0,2,0). The nearest-neighbor singlet model does not produce a signal contribution at (0,2,0), indicating that correlations beyond nearest neighbors (or additional interaction terms beyond J_1) are required.
(c) Low-energy additional broad peaks not explained by nearest-neighbor interactions.
In the low-energy window 0.25–1 meV, extra broad peaks appear around reciprocal-space positions such as (1,0,0) and symmetry-equivalent points (again using the convention of doi: 10.1038/nature11659). These are not Bragg peaks (no static order) and are inconsistent with a nearest-neighbor-only description of the kagome-plane dynamics, pointing to additional correlations/interactions shaping the low-energy response.
(d) Energy-insensitive continuum behavior across a wide window.
A similar hexagonal ring-like scattering pattern persists across roughly 1.5–11 meV, with the overall scattering characteristics changing very weakly with energy. This "energy-insensitive" continuum contrasts with conventional nearest-neighbor-dominated magnets, where one typically observes energy-sensitive, dispersive spin-wave features and strong energy dependence of the scattering signature.

2. Impact of beyond-nearest-neighbor correlations on the ground state spin liquid
The observed longer-range (non-nearest-neighbor) correlations modify both the stability and the excitation content of the ground state:
(a) Reinforcing the absence of static order and spin freezing.
The kagome lattice is geometrically frustrated, intrinsically suppressing long-range magnetic order. Importantly, correlations beyond nearest neighbors in this setting tend to **further suppress** local-moment ordering tendencies by enhancing competing quantum fluctuations and frustration pathways, helping stabilize a disordered QSL ground state. Consistent with this, INS observes no sharp spin-wave modes and no magnetic ordering even down to very low temperatures (reported as low as $T \approx 0.05$ K in the INS literature), supporting a ground state maintained by strong quantum fluctuations rather than conventional order.
(b) Supporting fractionalized excitations (spinon continuum).
INS detects a broad excitation continuum (not sharp magnon dispersion surfaces) over approximately the 2–11 meV range. Such a continuum is widely interpreted as evidence for fractionalized excitations (spinons) in a QSL, and the detailed momentum modulation of this continuum is sensitive to longer-range correlations and additional exchange terms that reshape the spinon dispersion and spectral-weight distribution.
(c) "Short-range correlations but long-range coherence."
A key qualitative motif for spin liquids is that static correlations can be short-ranged while the quantum state maintains long-range coherence/entanglement. Correlations extending beyond nearest neighbors can contribute to sustaining this coherence by delocalizing singlet/triplet fluctuations over larger loops and distances, thereby stabilizing an entangled ground state without condensing into static order.

3. Why Herbertsmithite does not conform to a simple short-range RVB picture, and challenges to theory
(a) Gaplessness contradicts the simplest short-range RVB expectation.
A simple short-range RVB state built primarily from random nearest-neighbor singlets is typically associated with a **gapped** spin liquid. Experimentally, however, no spin gap is observed across the measured momentum range down to energies as low as $\hbar\omega \approx 0.25$ meV. The persistence of low-energy spectral weight throughout Q–ω space is therefore inconsistent with a basic gapped short-range RVB description.
(b) Peak narrowing and special-momentum intensity require longer-range correlations.
A random nearest-neighbor singlet model would produce a broad, strongly diffuse $S(\mathbf{Q})$ with widths set by one-bond correlations. The experimentally observed narrowing of scattering features in reciprocal space, together with intensity at momenta such as (0,2,0) (doi: 10.1038/nature11659 convention), cannot be explained by purely short-range singlet coverings; instead, longer-range correlations are required to modulate the structure factor and redistribute spectral weight.
(c) Implication: the Hamiltonian must extend beyond the nearest-neighbor kagome Heisenberg model.
To reproduce the detailed $S(\mathbf{Q}, \omega)$, theoretical descriptions generally need to augment the nearest-neighbor Heisenberg model with additional terms such as further-neighbor exchanges (e.g., J_2, diagonal J_d across hexagons) and anisotropies allowed by lattice symmetry (e.g., Dzyaloshinskii–Moriya interactions). The INS mismatches therefore motivate reconstructing an effective kagome-plane Hamiltonian containing non-nearest-neighbor interactions.
(d) Challenges posed to theory.
The experimental combination of (i) a globally gapless continuum, (ii) weak energy dependence of the scattering pattern over a broad window, and (iii) momentum-space features implying longer-range correlations, raises several open issues:

- **Gapless spin-liquid mechanism:** How do longer-range correlations in a 2D kagome system shape the spinon dispersion so that a broadly gapless continuum persists across the full momentum range?
- **Quantifying "short-range correlations + long-range coherence":** Experiments suggest coexistence of limited static correlation length with long-range quantum coherence, but a quantitative coupling law between correlation range, entanglement measures (e.g., entanglement entropy), and excitation spectra remains unclear.
- **Model reconstruction and identifiability:** Multiple extended-Hamiltonian candidates can qualitatively match parts of the data; producing falsifiable, high-precision predictions for the full $S(\mathbf{Q}, \omega)$ remains difficult, especially once realistic longer-range terms are included.

F.5 Industry

Background:
You are an expert in machine learning, and your current research focuses on underwater image enhancement. You have designed an encoder–decoder architecture and trained the model using Mean Squared Error (MSE) as the loss function. During training, you observe that the MSE on the training set steadily decreases, and the MSE on the validation set also decreases accordingly. However, the subjective visual quality on the test set is poor: the enhanced images appear overly smooth and lack fine details. Moreover, the PSNR and SSIM metrics are significantly lower than those of state-of-the-art (SOTA) models.

Question:
Based on the characteristics of the task and the properties of the loss function, analyze the reasons that may have led to this phenomenon.

Rubrics

1. Analytical Reasoning
(+6) Provide an in-depth analysis of the model and clearly point out the fundamental tendency of the MSE loss to drive the network toward predicting the conditional mean of all possible solutions (i.e., the mean solution).
(+4) Explain in detail the concrete effects of this averaging behavior—specifically, how high-frequency details such as edges and textures are transformed into smooth low-frequency components—and explicitly identify this process as the key cause of the over-smoothing phenomenon in the output images.
(+6) Accurately identify the pixel-wise equal-weighting limitation of the MSE loss—namely, that all pixels are assigned identical importance—leading the model to preferentially optimize large-area background regions at the expense of small but critical detail regions.
(+6) Point out that the downsampling operations in the encoder–decoder architecture (such as convolutions with stride and pooling operations) inevitably cause information loss in the image representation.
(+5) Through analysis, clearly state that downsampling operations have an inherent attenuation effect on high-frequency information such as edges and textures.
(+6) Clearly explain the limitations of the decoder's upsampling process, which can only reconstruct images through interpolation based on low-frequency features and is unable to actively recover high-frequency details that have already been lost.
(+6) Explicitly point out that the model may overfit to specific scenes or pixel distributions in the training set, resulting in insufficient generalization performance on previously unseen test scenarios.
(+8) Mention potential issues with the ground-truth labels, including two specific cases: (1) a mismatch between synthetic ground truth and real-world scene characteristics; and (2) misalignment between the labels and the input images.
(-4) After identifying the task as an ill-posed problem, the answer fails to introduce the core mathematical concept for addressing ill-posedness—regularization—and does not discuss how different loss functions provide implicit or explicit regularization.
(-3) The discussion is limited to technical flaws in ground-truth acquisition (e.g., misalignment errors) and fails to reflect on the more fundamental question of "what constitutes ground truth" in underwater image enhancement.
(+4) The answer proposes a three-stage improvement roadmap, such as quick verification (e.g., switching to L1 loss), performance enhancement (e.g., introducing perceptual loss), and the pursuit of state-of-the-art techniques (e.g., GAN-based methods).
(+3) In response to the core attribution that "MSE leads to over-smoothing," the answer designs a simple yet logically rigorous simulation or ablation experiment to demonstrate empirical thinking. An ideal example: "To verify that pixel-wise equal weighting causes background-dominated optimization, the following controlled experiment can be designed: (1) baseline: train with standard MSE."

(+4) Critically point out that not only PSNR and SSIM have limitations, but even current "perceptual quality" evaluations rely on limited pretrained models (e.g., VGG), and further propose a more systematic, multi-dimensional evaluation framework tailored to underwater image enhancement.
(+3) When proposing improvement strategies (such as using more complex loss functions or architectures), explicitly acknowledge the increased computational cost and discuss feasible trade-off strategies based on application scenarios (e.g., mobile devices or real-time systems).
(-2) The proposed improvement methods are disconnected from the problem attribution and do not explain how the new approaches (e.g., alternative loss functions) mechanistically address MSE-induced over-smoothing and detail loss.
(+4) At the beginning of the answer, the core issue—namely, the MSE loss function—is explicitly identified, without prematurely expanding into excessive background information or parallel arguments.
(+3) Each sub-argument adopts a topic–comment structure, with the first sentence of each paragraph clearly stating the central claim of that paragraph.

2. Structure and Formatting
(-3) The answer merely lists points in a parallel manner, without a clear top-level structure or hierarchical expansion that reflects a coherent main thread (e.g., the progressive relationship among task, loss function, architecture, and data).

Response

The phenomenon you observe stems from the fundamental mismatch between the mathematical properties of Mean Squared Error (MSE) and the perceptual requirements of underwater image enhancement, compounded by the ill-posed nature of the restoration task. Here is a detailed analysis:

1. Regression-to-the-mean effect (primary cause)
Mathematically, minimizing MSE corresponds to estimating the conditional expectation $\hat{y} = \mathbb{E}[y|x]$. Underwater image enhancement is a highly ill-posed inverse problem where multiple plausible clean images (y) can correspond to a single degraded input (x) due to varying water types, turbidity levels, illumination conditions, and intrinsic ambiguities in color recovery. When the model minimizes MSE across this multi-modal distribution, it converges to the pixel-wise average of all possible valid solutions. This averaging process destructively interferes with high-frequency components (textures, edges, fine details), resulting in the observed over-smoothness and "hazy" appearance, as high-frequency details rarely align perfectly across different valid restoration hypotheses.

2. Frequency bias and texture penalty
MSE is dominated by low-frequency components (global color cast, overall brightness) because errors in these channels contribute more to the pixel-wise squared sum. Consequently, the model prioritizes correcting the global blue-green color distortion and mean intensity—where large errors heavily penalize the loss—while suppressing fine details. High-frequency textures (e.g., coral patterns, sediment granules, fish scales) contribute minimally to the total MSE when smoothed, so the network learns to sacrifice these details to minimize variance and noise, generating conservative, blurry predictions that lack the sharp local contrasts essential for realistic underwater scenes.

3. Perceptual-statistical disconnect
MSE assumes pixel-wise independence and treats all errors equally regardless of their structural context. However, the Human Visual System (HVS) is highly sensitive to local contrast, edge coherence, and semantic consistency—properties that MSE does not encode. A model trained with MSE minimizes point-wise deviations without regard for whether the resulting image appears natural or contains coherent textures. This explains why your validation MSE decreases steadily (indicating successful statistical regression) while subjective quality deteriorates: the model achieves the mathematical optimum (minimum variance estimator) that is perceptually suboptimal.

4. Architecture-bottleneck compounding
Your encoder-decoder architecture likely employs downsampling operations (pooling/strided convolutions) that discard high-frequency information during encoding. While skip connections (e.g., U-Net) can mitigate this, MSE-based training provides no incentive for the decoder to reconstruct high-frequency details lost in the bottleneck or to use the skip connections effectively for texture synthesis. Unlike adversarial or perceptual losses that explicitly penalize unrealistic texture statistics, MSE is satisfied by smooth

interpolations that "fill in" missing details with local averages, exacerbating the information loss inherent in the architecture.

5. Metric implications (PSNR/SSIM divergence)

Although PSNR is mathematically derived from MSE (PSNR $= 10 \log_{10}(\frac{\text{MAX}_I^2}{\text{MSE}})$), your model achieves low PSNR relative to SOTA because:

- **Suboptimal convergence:** The model converges to a high-MSE local minimum characterized by the blurry mean predictions described above. While training MSE decreases, the absolute converged value remains higher than what SOTA methods achieve through better inductive biases or combined loss functions.
- **SSIM specificity:** SSIM measures luminance, contrast, and structure correlations. By optimizing for pixel-wise similarity rather than structural correlation, your model destroys local contrast (variance) and structural coherence—core components of SSIM—resulting in poor scores despite pixel-wise intensity averages being roughly correct.

6. Domain-specific challenges

Underwater images exhibit wavelength-dependent attenuation and backscatter, creating spatially varying degradation where local neighborhoods require non-uniform restoration. MSE enforces a global L_2 penalty that favors spatially uniform solutions, preventing the model from learning the aggressive local contrast enhancements needed to reverse scattering effects, further contributing to the flat, washed-out appearance. In summary, MSE is intrinsically unsuitable for this task because it seeks the statistically average solution to an ambiguous inverse problem, systematically erasing high-frequency details that define image quality, while providing no mechanism to enforce structural or perceptual plausibility.

G Limitation

G.1 Dynamic Expansion and Scenario Deepening

While the current five domains are representative, other numerous high-value fields (*e.g.*, energy, climate science, public policy) remain uncovered. Future work could incrementally incorporate more vertical domains and further refine sub-domains to enhance the benchmark's comprehensiveness and granularity. Beyond, building a live, evolving benchmark with the dynamic nature of real-world expertise by integrating real-time or frequently updated information will be the trend in the long run.

G.2 Automation of Fine-grained Process Evaluation

Although we improve rubric reliability through strict guidelines, verification, and consistency with human review, current rubrics are less objective than the equivalency checker of a single expression or number, and relies on the model judge's capabilities. Furthermore, although structured rubrics designed by experts are employed, full reliance on manual scoring is difficult to scale. Future research could develop more advanced automated evaluation mechanisms, such as rule-based or learning-based methods to automatically score intermediate steps like reasoning chains, evidence citation, and compliance checks, thereby improving efficiency while maintaining reliability.

As a starting point, $OneMillion-Bench aims to propel language agents towards greater professionalism, reliability, and practical utility. We hope $OneMillion-Bench can serve as a practical, challenging evaluation target for the next generation of language agents, systems expected not only to answer, but to *do the work* necessary to answer well. We look forward to collaborating with the community to iteratively improve the benchmark itself and to deeply explore the open questions outlined above, ultimately contributing to the safe, effective, and responsible deployment of AI in critical professional domains.

H Ethical Considerations

We adhere to ethical principles to ensure the responsible development and application of our proposed techniques. The research conducted in the paper conforms in every respect, with the NeurIPS code of ethics guidelines. We advocate for rigorous validation of tasks against diverse domains and sources in avoid of human biases. We also acknowledge the potential broader impacts of our research, recognize the environmental and computational costs associated with manually curated tasks and LLM evaluations, and strive to optimize our pipeline for higher efficiency and lower costs.

Table 7: Sampling parameters adopted in the models evaluation against the $OneMillion-Bench.

Model	Search Provider	Thinking Effort	Misc. Parameters
DeepSeek-V3.2	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 65,536
DeepSeek-V3.2-Speciale	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 65,536
MiniMax-M2.1	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 65,536
MiniMax-M2.5	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 196,608
Qwen3-235B-A22B-2507	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 262,100
Hunyuan-2.0-Thinking	Tencent	[On by default]	Temperature = 0 Max new tokens = 32,768
Step-3.5-Flash	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 256,000
Doubao-Seed-2.0-Pro-260215	ByteDance	High	Temperature = 0 Max new tokens = 128,000
Doubao-Seed-1.8-251228	ByteDance	High	Temperature = 0 Max new tokens = 64,000
GLM-4.7	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 131,072
GLM-5	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 131,072
Kimi-K2-Thinking	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 262,144
Kimi-K2.5	OpenRouter	High	Temperature = 0 Max new tokens = 262,144
GPT-5.4-High	OpenAI	High	Temperature = 0 Max new tokens = 128,000
GPT-5.2-High	OpenAI	High	Temperature = 0 Max new tokens = 128,000
GPT-5.3-Codex	OpenAI	High	Temperature = 0 Max new tokens = 128,000
Claude-Opus-4.5	Anthropic	High	Temperature = 0 Max new tokens = 64,000
Claude-Opus-4.6	Anthropic	High	Temperature = 0 Max new tokens = 128,000
Claude-Sonnet-4.5	Anthropic	High	Temperature = 0 Max new tokens = 64,000
Claude-Sonnet-4.5	Anthropic	High	Temperature = 0 Max new tokens = 64,000
Gemini-3-Pro-preview	Google	High	Temperature = 0 Max new tokens = 65,500
Grok-4	xAI	[On by default]	Temperature = 0 Max new tokens = 256,000
Qwen3-Max-2026-01-23	Alibaba	High	Temperature = 0 Max new tokens = 32,768
Qwen3.5-Plus-2026-02-15	Alibaba	High	Temperature = 0 Max new tokens = 65,536

Table 8: Comparison of Expert Score across different domains on the **Global evaluation set**.

Model / Agentic System	Finance		Healthcare		Industry		Law		Natural Science	
	Vanilla	Search	Vanilla	Search	Vanilla	Search	Vanilla	Search	Vanilla	Search
GPT-5.4-HIGH	42.0	68.8 ↑26.8	63.6	53.9 ↓9.7	54.7	58.2 ↑3.5	55.2	55.6 ↑0.5	59.4	59.4 ↓0.0
CLAUDE-OPUS-4.6	42.1	55.5 ↑13.4	68.4	73.7 ↑5.3	58.9	67.5 ↑8.7	53.0	59.5 ↑6.5	52.4	58.9 ↑6.5
GPT-5.3-CODEX	36.2	63.2 ↑27.1	53.2	48.8 ↓4.4	50.4	48.2 ↓2.1	46.7	50.8 ↑4.0	53.8	42.0 ↓11.9
QWEN3.5-PLUS	30.7	31.3 ↑0.6	59.3	60.9 ↑1.6	50.8	52.8 ↑2.0	49.7	48.6 ↓1.1	47.5	51.1 ↑3.6
GPT-5.2-HIGH	29.3	60.7 ↑31.4	59.5	59.3 ↓0.2	50.1	52.5 ↑2.4	48.8	57.0 ↑8.2	50.0	54.7 ↑4.7
KIMI-K2.5	34.6	31.6 ↓3.0	54.8	48.1 ↓6.7	46.0	44.4 ↓1.5	43.8	36.9 ↓6.8	46.4	44.8 ↓1.6
GEMINI-3-PRO-PREVIEW	35.8	47.6 ↑11.8	52.3	61.7 ↑9.3	42.0	50.4 ↑8.4	47.7	50.7 ↑3.0	40.6	53.6 ↑13.0
DEEPSEEK-V3.2-SPECIALE	27.6	28.5 ↑1.0	48.9	45.7 ↓3.1	43.2	37.1 ↓6.1	38.3	36.5 ↓1.8	44.3	45.3 ↑1.0
DOUBAO-SEED-2.0-PRO	25.9	35.8 ↑9.9	51.7	64.3 ↑12.6	38.9	52.6 ↑13.7	35.4	55.0 ↑19.6	45.2	51.3 ↑6.1
GROK-4	27.3	51.4 ↑24.2	45.3	47.5 ↑2.1	39.0	38.5 ↓0.5	39.5	46.4 ↑6.8	42.0	45.6 ↑3.6
GLM-5	30.9	34.5 ↑3.7	47.7	51.9 ↑4.2	36.0	39.1 ↑3.1	38.0	39.6 ↑1.6	39.9	40.5 ↑0.6
MINIMAX-M2.1	19.9	25.4 ↑5.5	46.7	41.4 ↓5.2	39.6	34.6 ↓5.0	31.4	30.8 ↓0.6	36.7	37.3 ↑0.6
HUNYUAN-2.0	25.0	22.6 ↓2.4	46.4	36.7 ↓9.8	34.6	38.3 ↑3.8	28.7	24.9 ↓3.8	38.7	28.4 ↓10.3
STEP-3.5-FLASH	23.5	25.9 ↑2.4	45.3	43.6 ↓1.7	37.9	27.8 ↓10.1	29.2	29.1 ↓0.1	36.0	38.3 ↑2.4
LING-2.5-1T	24.2	22.9 ↓1.3	43.5	40.8 ↓2.7	37.9	36.9 ↓1.0	23.8	25.0 ↑1.1	36.2	30.5 ↓5.7
MINIMAX-M2.5	19.9	16.7 ↓3.3	33.1	38.8 ↑5.7	36.4	39.3 ↑2.9	27.5	26.5 ↓1.0	30.5	36.2 ↑5.7
Deep Research Agents										
O3-DEEPRESEARCH	49.4		50.9		43.9		50.7		36.5	
SONAR-DEEPRESEARCH	45.5		42.7		43.8		41.6		38.0	
O4-MINI-DEEPRESEARCH	40.3		40.3		34.3		36.4		37.1	

Table 9: Comparison of Expert Score across different domains on the **CN evaluation set**.

Model / Agentic System	Finance		Healthcare		Industry		Law		Natural Science	
	Vanilla	Search	Vanilla	Search	Vanilla	Search	Vanilla	Search	Vanilla	Search
GPT-5.4-HIGH	53.6	63.3 ↑9.7	67.3	57.1 ↓10.1	55.8	57.5 ↑1.7	43.9	53.6 ↑9.7	60.6	62.1 ↑1.5
CLAUDE-OPUS-4.6	46.9	60.9 ↑14.0	68.4	70.6 ↑2.2	59.0	69.7 ↑10.7	49.4	60.9 ↑11.4	55.5	60.7 ↑5.2
QWEN3.5-PLUS	48.4	46.9 ↓1.4	60.1	61.7 ↑1.5	50.2	49.0 ↓1.2	48.9	56.1 ↑7.2	49.6	46.7 ↓2.9
GPT-5.2-HIGH	41.4	56.1 ↑14.8	57.0	60.8 ↑3.8	46.4	50.3 ↑4.0	35.0	42.5 ↑7.4	55.8	53.3 ↓2.5
GPT-5.3-CODEX	45.6	54.7 ↑9.1	57.1	54.9 ↓2.2	45.6	54.6 ↑9.0	37.0	41.2 ↑4.3	49.1	49.2 ↑0.1
GEMINI-3-PRO-PREVIEW	46.4	56.4 ↑10.0	53.9	60.3 ↑6.5	42.5	50.9 ↑8.4	40.5	54.7 ↑14.2	46.7	47.6 ↑0.9
KIMI-K2.5	48.8	42.5 ↓6.3	53.0	46.6 ↓6.5	39.2	46.9 ↑7.6	40.3	40.1 ↓0.3	45.4	42.3 ↓3.1
DEEPSEEK-V3.2-SPECIALE	37.1	44.0 ↑6.9	49.9	49.5 ↓0.3	39.6	42.1 ↑2.5	31.0	43.2 ↑12.1	48.1	43.1 ↓5.0
GLM-5	39.5	37.4 ↓2.1	48.5	48.3 ↓0.1	38.3	38.6 ↑0.2	37.1	39.2 ↑2.1	40.2	42.1 ↑1.9
DOUBAO-SEED-2.0-PRO	36.6	48.6 ↑12.0	43.9	54.4 ↑10.6	43.7	50.7 ↑7.0	37.8	51.3 ↑13.5	41.3	47.4 ↑6.1
GROK-4	37.3	43.5 ↑6.2	45.5	45.2 ↓0.3	39.0	36.8 ↓2.2	35.9	30.7 ↓5.2	41.1	41.8 ↑0.6
LING-2.5-1T	29.5	33.6 ↑4.1	46.4	42.8 ↓3.6	40.3	38.9 ↓1.4	31.6	26.3 ↓5.4	35.8	36.4 ↑0.5
HUNYUAN-2.0	36.1	36.6 ↑0.5	37.1	33.2 ↓3.9	41.1	34.1 ↓7.0	35.2	32.1 ↓3.2	32.0	28.3 ↓3.7
STEP-3.5-FLASH	38.5	33.8 ↓4.7	44.2	50.1 ↑5.9	36.1	30.3 ↓5.8	28.4	28.0 ↓0.4	33.8	36.1 ↑2.3
MINIMAX-M2.1	31.6	35.9 ↑4.4	46.2	45.1 ↓1.0	43.3	38.0 ↓5.3	27.6	30.2 ↑2.6	30.2	34.4 ↑4.2
MINIMAX-M2.5	29.4	29.8 ↑0.4	40.3	34.2 ↓6.1	39.4	34.8 ↓4.6	25.0	25.2 ↑0.1	33.0	36.5 ↑3.5
Deep Research Agents										
O3-DEEPRESEARCH	47.4		46.4		40.1		34.9		39.5	
SONAR-DEEPRESEARCH	39.1		42.0		43.1		39.2		37.7	
O4-MINI-DEEPRESEARCH	39.5		39.8		34.7		27.7		38.3	

Table 10: Economic value ($USD) of $OneMillion-Bench by domain and regional subsets.

Domain	Count	Subset	Avg. Time (h)	Avg. Wage (/h)	Avg. Value/Q	Total Value
Finance	80	CN (¥)	23.5	312.6	7,410.8	296,432
		Global ($)	26.1	175.7	4,593.2	183,726
Law	80	CN (¥)	26.2	180.4	4,722.5	188,901
		Global ($)	31.4	243.0	7,666.1	306,646
Healthcare	80	CN (¥)	22.9	189.9	4,379.9	175,197
		Global ($)	22.9	354.9	8,188.9	327,557
Natural Science	80	CN (¥)	22.4	118.5	2,687.3	107,493
		Global ($)	22.4	84.6	1,907.0	76,280
Industry	80	CN (¥)	24.5	152.1	3,845.2	153,809
		Global ($)	24.5	114.9	2,854.0	114,161
Total (Total Value)					CN (¥)	**921,832**
					Global ($)	**1,008,370**